

Card column reference

All **711** columns across the five cards, grouped by card and then by block, with blocks ordered by **measured importance**. Each entry leads with what the column *is*; the caveats fold underneath.

Generated 2026-08-20 by `analyses/ops/gen_column_reference.py` from `card_glossary.tsv` and `card_registry.tsv`. **Do not hand-edit**.

Vintage mix “ read before quoting a fitted column

The delivered edge card is **16 days newer** than the base it layers over. Its **84 base-owned columns** (`coupling_`, `beta_`, `dose_`, `share_`, `rank_`) carry the older vintage; the other 102 are current.

The one rule behind this layout

The rung is a property of (CARD, COLUMN) “ never of the column name. `beta` is edge-rung on the edge card and family-rung on the `gene_family` card: same estimator, different unit. That is why this is organised card-first, and why one name can appear twice with different text.

How blocks are ordered

By three measured signals, not by size: how often a block's columns are named in the **discovery registry** (what the findings rest on, weighted highest), how many **modules** read them, and whether the block is **base-owned** (fitted) rather than an annotation join. Names under 6 characters count only backtick-quoted registry hits “ otherwise `n` and `z` match ordinary prose.

One correction the measurement needed: mention-frequency is not importance “ a *settled* block gets discussed less precisely because nobody argues about it. `adm_` scored 27th of 44 on the raw signals while MH-216 calls `adm_has_site` the single most load-bearing conditioning variable, on 4 registry mentions. Five blocks therefore carry a **curated floor**, each citing the finding that justifies it (see `CURATED_FLOOR`). That is the only hand-set input here.

Cards

- edge “ 186 columns, 47 blocks
- gene “ 135 columns, 29 blocks
- gene_family “ 61 columns, 12 blocks
- arm “ 296 columns, 44 blocks
- seed_family “ 33 columns, 2 blocks

edge

One row per one (miRNA arm, gene) pair. The widest card and the one most claims are made from. 84 of its columns come from the BASE card and 102 from the annotation layer “ check the vintage banner before quoting a fitted one.

186 columns 5,649 rows 47 blocks median coverage 72%

Contents “ 47 blocks, most important first

1. (bare) 6 “ CORE

2. coupling_ 13 “ CORE

3. seed_ 1 “ CORE

4. arm_ 15 “ CORE

5. adm_ 4 “ CORE

6. beta_ 8 “ CORE

7. n_ 5 “ CORE

8. share_ 4 “ CORE

9. shift_ 1 “ CORE

10. cal_ 7 “ CORE

11. echim_ 3 “ CORE

12. rank_ 4 “ CORE

13. healthy_ 3 “ SUPPORTING

14. ctx_ 18 “ SUPPORTING

15. identity_ 5 “ SUPPORTING

16. dose_ 7 “ SUPPORTING

17. d_rank_ 4 “ SUPPORTING

18. gene_ 6 “ SUPPORTING

19. ago_ 2 “ SUPPORTING

20. pip_ 2 “ SUPPORTING

21. retention_ 3 “ REFERENCE

22. oof_ 3 “ REFERENCE

23. grank_ 3 “ REFERENCE
24. composition_ 1 “ REFERENCE

25. w_ 1 “ REFERENCE

26. own_ 1 “ REFERENCE

27. dGlobal_ 3 “ REFERENCE

28. m_ 1 “ REFERENCE

29. term_ 3 “ REFERENCE

30. prior_ 1 “ REFERENCE

31. edge_ 1 “ REFERENCE

32. mean_ 2 “ REFERENCE

33. surrogate_ 2 “ REFERENCE

34. esub_ 8 “ REFERENCE

35. realization_ 2 “ REFERENCE

36. kd_ 2 “ REFERENCE

37. wiring_ 1 “ REFERENCE

38. cell_ 1 “ REFERENCE

39. sd_ 1 “ REFERENCE

40. z_ 1 “ REFERENCE

41. acquisition_ 1 “ REFERENCE

42. dShare_ 2 “ REFERENCE

43. heur_ 1 “ REFERENCE

44. family_ 1 “ REFERENCE

45. cptac_ 20 “ REFERENCE

46. hly_ 1 “ REFERENCE

47. is_ 1 “ REFERENCE

(BARE) “ 6

core “ the columns findings rest on

6 columns “ spans edge, key “ this prefix groups columns living on DIFFERENT units “ coverage 96%“100% (median 100%)

2× identifier “ 2× real “ 0 “ 1× boolean “ 1× real, signed

arm

KEY

IDENTIFIER

100%

KEY “ mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()` , which gates against the canonical universe and LOGS drops rather than silently dropping.

AUTHORED

beta

EDGE

REAL “ 0

100%

EDGE-rung coupling coefficient “ `readouts.run(level='arm')` , the same-seed collapse REMOVED.

AUTHORED

gene	KEY	IDENTIFIER	100%	KEY “ HGNC gene symbol. AUTHORED
identified	EDGE	BOOLEAN	100%	$ z > 2$ on the UNCALIBRATED posterior SD, where $z = \text{beta} / \text{beta_sd}$ (MH-83 records precision 0.86 / recall 0.89 against a held-out standard). True on 24.8% of edges. AUTHORED
identity	EDGE	REAL, SIGNED	96%	Shapley/LMG share of the gene's R^2 credited to this edge, with bagged-NNLS weights. AUTHORED
z	EDGE	REAL $\hat{\beta} / \hat{\sigma}$	100%	$\text{beta} / \text{beta_sd}$ for the EDGE-level fit (<code>readouts.run(level='arm')</code>), the whole gene's arms as separate predictors), on the UNCALIBRATED posterior SD. AUTHORED

COUPLING_ 13

core “ the columns findings rest on

Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.

13 columns $\hat{\beta}$ spans edge, family “ this prefix groups columns living on DIFFERENT units $\hat{\beta}$ coverage **3%**“**100%** (median 62%) $\hat{\beta}$ 9 share the block description

5x signed $\hat{\beta}^{\prime 1} \hat{\beta} \{1$ $\hat{\beta}$ 5x p-value $\hat{\beta}$ 3x real, signed

coupling_fam	FAMILY	SIGNED $\hat{\beta}^{\prime 1} \hat{\beta} \{1$	100%	Family-level coupling for the edge's (gene, seed_family) cell. AUTHORED
coupling_hly	EDGE	SIGNED $\hat{\beta}^{\prime 1} \hat{\beta} \{1$	56%	in the GTEx-healthy state DERIVED <i>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state “ many arms are multi-mapping-collapsed, MH-166)</i>
coupling_hly_surrogate	EDGE	SIGNED $\hat{\beta}^{\prime 1} \hat{\beta} \{1$	3%	Healthy coupling via a same-seed SURROGATE arm, used where the arm itself has no GTEx signal. AUTHORED
coupling_nat	EDGE	SIGNED $\hat{\beta}^{\prime 1} \hat{\beta} \{1$	62%	in matched normal (NAT) DERIVED <i>Defined on: edges with a matched NAT leg (n~104 paired participants)</i>
coupling_p_hly	EDGE	P-VALUE	56%	p-value $\hat{\beta}$ in the GTEx-healthy state DERIVED <i>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state “ many arms are multi-mapping-collapsed, MH-166)</i>
coupling_p_hly_resolved	EDGE	P-VALUE	59%	p-value $\hat{\beta}$ in the GTEx-healthy state DERIVED <i>Defined on: arms with a same-seed SURROGATE (MH-166) “ 3.6% of edges</i>
coupling_p_hly_surrogate	EDGE	P-VALUE	3%	p-value $\hat{\beta}$ in the GTEx-healthy state DERIVED <i>Defined on: arms with a same-seed SURROGATE (MH-166) “ 3.6% of edges</i>
coupling_p_nat	EDGE	P-VALUE	62%	p-value $\hat{\beta}$ in matched normal (NAT) DERIVED <i>Defined on: edges with a matched NAT leg (n~104 paired participants)</i>
coupling_p_tum	EDGE	P-VALUE	69%	p against the CALIBRATED site-free null (not a theoretical t-null). Median 0.375; <0.05 on 16.2%. AUTHORED
coupling_tum	EDGE	SIGNED $\hat{\beta}^{\prime 1} \hat{\beta} \{1$	72%	in tumour DERIVED <i>Defined on: edges present in the progression panel (arm expressed + state measured)</i>

coupling_z_hly EDGE REAL, SIGNED59%	z-score $\hat{\mu}$ in the GTEx-healthy state DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state $\hat{\mu}$ many arms are multi-mapping-collapsed, MH-166)
coupling_z_nat EDGE REAL, SIGNED62%	z-score $\hat{\mu}$ in matched normal (NAT) DERIVED Defined on: edges with a matched NAT leg ($n \sim 104$ paired participants)
coupling_z_tum EDGE REAL, SIGNED69%	The tumour coupling expressed in NULL SDs $\hat{\mu}$ the CALIBRATED effect size, and the scale to quote. Median $\hat{\mu}^*0.32$; $z > 2$ on 15.5%, $z > 3$ on 6.4%. AUTHORED
SEED_ $\hat{\mu}$ 1 core $\hat{\mu}$ the columns findings rest on	
1 columns $\hat{\mu}$ rung family $\hat{\mu}$ coverage 100%$\hat{\mu}$100% (median 100%) 1x identifier	
seed_family FAMILY IDENTIFIER100%	KEY $\hat{\mu}$ the seed family itself, one row per family. AUTHORED
ARM_ $\hat{\mu}$ 15 core $\hat{\mu}$ the columns findings rest on	
15 columns $\hat{\mu}$ spans arm, arm-in-family, family $\hat{\mu}$ this prefix groups columns living on DIFFERENT units $\hat{\mu}$ coverage 20%$\hat{\mu}$100% (median 69%) 4x real $\hat{\mu} \geq 0$ $\hat{\mu}$ 4x real, signed $\hat{\mu}$ 3x fraction $0 \leq 1$ $\hat{\mu}$ 2x categorical $\hat{\mu}$ 1x percent $\hat{\mu}$ 1x boolean	
arm_credit_share ARM-IN-FAMILY FRACTION $0 \leq 1$20%	This arm's share of its (gene, seed_family) cell's credit. $\hat{\mu}$... Verified to sum to 1.0000 within every one of the 467 multi-arm cells, and it genuinely varies PER ARM inside the cell (467/467). AUTHORED Defined on: multi-arm cells only ($n_{arm_in_cell} > 1$) $\hat{\mu}$ 20.3% of edges
arm_dbeta FAMILY REAL $\hat{\mu} \geq 0$20%	$\max \hat{\mu}^2 - \min \hat{\mu}^2$ across the cell's member arms. AUTHORED
arm_detection ARM FRACTION $0 \leq 1$95%	Fraction of patients in which this ARM is measured at all (readouts._detection). AUTHORED
arm_id_status ARM-IN-FAMILY CATEGORICAL100%	How the arm's identity within its cell was resolved: resolved_by_dose 2,344 $\hat{\mu}$ singleton 2,260 $\hat{\mu}$ unvalidated_balanced 1,045. AUTHORED
arm_iqr ARM REAL $\hat{\mu} \geq 0$69%	Interquartile range of the arm's abundance $\hat{\mu}$ its DYNAMIC RANGE. AUTHORED
arm_lfc_HLY_NAT_raw ARM REAL, SIGNED63%	log-FC GTEx-healthy $\hat{\mu}$ TCGA-NAT, raw. AUTHORED
arm_lfc_HLY_TUM_QN ARM REAL, SIGNED64%	log-FC GTEx-healthy $\hat{\mu}$ TCGA-tumour, on quantile-normalised values. AUTHORED
arm_lfc_HLY_TUM_raw ARM REAL, SIGNED63%	As arm_lfc_HLY_TUM_QN but against the RAW GTEx median. AUTHORED

arm_lfc_NAT_TUM ARMREAL, SIGNED68%	log-FC of the arm's median abundance, matched NAT â†’ tumour. AUTHORED
arm_med_rpm ARMREAL 069%	Median linear RPM of this ARM across the cohort. Arm-rung, so it REPEATS across the arm's genes. Fill 0.686 â€” the gap is arms with no expression record, not zeros. AUTHORED Defined on: edges present in the progression panel (arm expressed + state measured)
arm_pct_above_floor ARMPERCENT69%	Percent of samples in which the arm sits above the detection floor (median 99). Feeds arm_spiker together with arm_iqr. AUTHORED Defined on: edges present in the progression panel (arm expressed + state measured)
arm_role_in_family ARM-IN-FAMILYCATEGORICAL72%	â€” The ARM's DOSE role inside its seed family â€” NOT the family's role in the gene, which is what the name suggests. AUTHORED
arm_sep_z FAMILYREAL 020%	arm_dbeta in units of the cell's own pooled posterior SD â€” can these same-seed arms be told apart at all. AUTHORED
arm_share_of_family_dose ARM-IN-FAMILYFRACTION 0â€”169%	The arm's share of its seed family's TOTAL dose â€” denominator is the full family , not the design (median 0.69). The quantity family_role is cut from. AUTHORED Defined on: edges present in the progression panel (arm expressed + state measured)
arm_spiker ARMBOOLEAN69%	arm_pct_floor < 40 AND arm_iqr > 1.5 â€” the arm sits at/below the detection floor in most samples yet swings widely when present. AUTHORED

ADM_ 4

core â€” the columns findings rest on

4 columns â€” rung edge â€” per-edge FLAGS; the arm card carries their rollup â€” coverage 87%â€”87% (median 87%) 4x boolean
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adm_admissible EDGEBOOLEAN87%	The admissibility verdict: site + evidence + expression. AUTHORED
adm_expr_gtex EDGEBOOLEAN87%	ADMISSIBILITY â€” could this edge repress in breast AT ALL? has_site (seed match in the 3'UTR) AND expressed (arm present in TCGA tumour OR GTEx healthy breast). AUTHORED
adm_expr_tcg EDGEBOOLEAN87%	Is the arm expressed above floor in TCGA? AUTHORED
adm_has_site EDGEBOOLEAN87%	Does the arm have a seed match in this target's 3'UTR. AUTHORED

BETA_ 8

core â€” the columns findings rest on

8 columns â€” rung edge â€” coverage 20%â€”100% (median 100%) 4x real 0 â€” 4x fraction 0â€”1

beta_arm EDGEREAL 020%	The arm's OWN Î² from the within-cell fit (arm_rung.py), defined only in multi-arm cells (fill 0.203). AUTHORED
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beta_deconv EDGE REAL $\hat{\beta}_{\text{deconv}} \approx 0$ 100%	$\hat{\beta}^2$ refit on the DECONV design (core C + the 8 non-malignant Wu-major lineages, Cancer-Epithelial held out). AUTHORED
beta_frac EDGE FRACTION $\hat{\beta}_{\text{frac}} \approx 1$ 100%	MAGNITUDE share (beta_f / sum beta). AUTHORED
beta_frac_abs EDGE FRACTION $\hat{\beta}_{\text{frac_abs}} \approx 1$ 100%	$ E[\hat{\beta}^2_{\text{f}}] / \hat{E}[E[\hat{\beta}^2]] \approx$ the share computed from the POINT ESTIMATES. AUTHORED
beta_frac_deconv EDGE FRACTION $\hat{\beta}_{\text{frac_deconv}} \approx 1$ 100%	beta_frac recomputed on the deconv design $\approx$ the composition-adjusted share of the gene's budget. AUTHORED
beta_frac_sd EDGE FRACTION $\hat{\beta}_{\text{frac_sd}} \approx 1$ 100%	Posterior SD of the per-draw fraction $\hat{\beta}^2_{\text{f}} / \hat{E}[\hat{\beta}^2]$. AUTHORED
beta_sd EDGE REAL $\hat{\beta}_{\text{sd}} \approx 0$ 100%	Posterior summaries of the coupling coefficient from the dense Gibbs fit (learned/readouts.py). _sd is the posterior SD, _frac the share of the gene's total beta. AUTHORED
beta_sd_deconv EDGE REAL $\hat{\beta}_{\text{sd_deconv}} \approx 0$ 100%	Posterior SD on the deconv design. AUTHORED

N_ $\hat{\beta} \cdot 5$
core $\approx$ the columns findings rest on

5 columns $\hat{\beta} \cdot$ **spans edge, family, gene** $\approx$ this prefix groups columns living on DIFFERENT units $\hat{\beta} \cdot$ coverage **20% $\approx$ 100%** (median 100%)
4 $\times$ count $\hat{\beta} \cdot$ 1 $\times$ constant

n_HLY_meas EDGE COUNT 72%	How many of this gene's arms have a MEASURED healthy (GTEx) level $\approx$ as opposed to one manufactured by the multi-mapping collapse. AUTHORED
n_arm_in_cell FAMILY COUNT 20%	How many arms sit in this edge's (gene, seed_family) CELL $\approx$ i.e. how many of that family's members appear in THIS gene's design. AUTHORED
n_fam GENE COUNT 100%	How many families compete at this edge's GENE. Gene-rung $\approx$ it repeats across the gene's edges by construction. beta is mathematically inert where it is 1. AUTHORED
n_family_in_design FAMILY COUNT 100%	Members of this arm's seed family THAT ARE IN THIS GENE'S DESIGN $\approx$ 1 on 4,497 of 5,644 edges (79.7%) . AUTHORED
n_samples GENE CONSTANT 100%	Samples the edge was fit on $\approx$ CONSTANT (1,040) across the card. AUTHORED

SHARE_ $\hat{\beta} \cdot 4$
core $\approx$ the columns findings rest on

The arm's share of the gene's total regulatory dose in one state (HLY / NAT / TUM). Within-gene, sums to 1 across the gene's arms.
4 columns $\hat{\beta} \cdot$ rung **edge** $\hat{\beta} \cdot$ coverage **46% $\approx$ 72%** (median 72%) $\hat{\beta} \cdot$ **2** share the block description
4 $\times$ fraction $\hat{\beta} \approx 1$

share_HLY EDGE FRACTION $\hat{\beta}_{\text{share_HLY}} \approx 1$ 72%	The arm's share of the gene's healthy (GTEx) pressure budget, median 0.129 . AUTHORED
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share_HLY_meas	share_HLY over ONLY the arms GTEx genuinely measures “invisible arms ABSTAIN (NaN) instead of being zeroed (MH-210). Median 0.244 against the imputed 0.129. AUTHORED
EDGE FRACTION 0-1 46%	
share_NAT	in matched normal (NAT) DERIVED
EDGE FRACTION 0-1 72%	<i>Defined on: edges with a matched NAT leg (n~104 paired participants)</i>
share_TUM	in tumour DERIVED
EDGE FRACTION 0-1 72%	<i>Defined on: edges present in the progression panel (arm expressed + state measured)</i>

SHIFT_ 1
core “the columns findings rest on

1 columns run edge coverage 72-72% (median 72%) 1x categorical

shift_class	Cross-state class for the edge. AUTHORED
EDGE CATEGORICAL 72%	

CAL_ 7
core “the columns findings rest on

CALIBRATED posterior width (learned/calibration.py): the reported SD corrected for the measured 1.18-1.34x narrowness, with _lo/_hi spanning the correction constant's own +/-1 SD. 7 columns run edge coverage 100-100% (median 100%) 4 share the block description 4x real 0 3x boolean
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cal_beta_sd	standard deviation DERIVED
EDGE REAL 0 100%	
cal_beta_sd_hi	standard deviation DERIVED
EDGE REAL 0 100%	
cal_beta_sd_lo	standard deviation DERIVED
EDGE REAL 0 100%	
cal_identified	Is the edge identified (cal_z > 2) under the CALIBRATED posterior width “the honest version of identified : 1,110 vs 1,375 edges , i.e. 19.9% [18.1-22.4] against 24.8%. AUTHORED
EDGE BOOLEAN 100%	
cal_identified_hi	The UPPER end of the calibrated identification band “ cal_z > 2 under the optimistic calibration (990 edges vs 1,110 central). AUTHORED
EDGE BOOLEAN 100%	
cal_identified_lo	The LOWER end of the calibrated identification band (1,234 edges vs 1,110 central). See cal_identified_hi “the pair is the uncertainty, and reporting one alone hides it. AUTHORED
EDGE BOOLEAN 100%	
cal_z	z-score DERIVED
EDGE REAL 0 100%	

ECHIM_ 3

core the columns findings rest on

3 columns

rung

edge

coverage

8%26%

(median 22%)

2x count

1x real

0

echim_manakov_w

Manakov chimeric-eCLIP support weight for this edge (median 3, max 1,712).

AUTHORED

EDGE

REAL

0

22%

echim_n_sources

How many chimeric-eCLIP sources support this edge (1 on 1,151 2 on 262 3 on 34).

AUTHORED

EDGE

COUNT

26%

echim_tarbase_w

TarBase chimeric support weight (median 2, max 28) a much shorter tail than the Manakov twin, and defined on only 8.4% of edges.

AUTHORED

EDGE

COUNT

8%

Defined on: edges with a chimeric (AGO-ligation) duplex as per-source, NEVER pooled; cross-tissue, not breast

RANK_ 4

core the columns findings rest on

The arm's WITHIN-GENE rank by dose in one state.

4 columns

rung

edge

coverage

46%72%

(median 72%)

4 share the block description

4x count

rank_HLY

in the GTEx-healthy state

DERIVED

EDGE

COUNT

72%

Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)

rank_HLY_meas

in the GTEx-healthy state measured only (un-imputed)

DERIVED

EDGE

COUNT

46%

Defined on: edges whose gene has 1 GTEx-MEASURED regulator as NaN = GTEx cannot see it (abstention), NOT 0

rank_NAT

in matched normal (NAT)

DERIVED

EDGE

COUNT

72%

Defined on: edges with a matched NAT leg (n~104 paired participants)

rank_TUM

in tumour

DERIVED

EDGE

COUNT

72%

Defined on: edges present in the progression panel (arm expressed + state measured)

HEALTHY_ 3

supporting read these when the core raises a question

3 columns

rung

arm

identical column set to the arm block

coverage

64%72%

(median 64%)

1x categorical

1x real

0

1x boolean

healthy_leg

Provenance of this arm's GTEx healthy leg measured , collapse_blind or true_absent.

AUTHORED

ARM

CATEGORICAL

72%

healthy_potential

The arm's IMPUTED healthy abundance a repression-POTENTIAL proxy, collapse-repaired (0.27 17.9).

AUTHORED

ARM

REAL

0

64%

healthy_uninformative

Is the healthy leg both collapse_blind AND high-potential i.e. the one combination where an *acquired* call is genuinely unsafe? True on 143 of 3,490 edges (4%).

AUTHORED

ARM

BOOLEAN

64%

CTX_ 18

supporting “ read these when the core raises a question

Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

18 columns spans arm, gene “ this prefix groups columns living on DIFFERENT units coverage 84%“100% (median 99%) 10 share the block description

4x real 0 4x signed 1 3x real, signed 3x count 2x boolean 1x categorical 1x fraction 0“1

ctx_apriori_class GENE CATEGORICAL 100%	class label DERIVED
ctx_arm_abundant ARM BOOLEAN 95%	ctx_arm_dose >= log2(11) “ the arm-abundance gate, ARM-rung on an edge card. See the arm card's entry. AUTHORED Defined on: arms present on the EDGE card (729 of 2,450) “ lifted arm-rung columns, names preserved
ctx_arm_dose ARM REAL 0 95%	This arm's median log2(RPM+1) “ see the arm card's entry. AUTHORED
ctx_ceiling GENE SIGNED 1 99%	The UPPER BOUND on what ANY estimator can achieve for this gene “ the 5-fold out-of-fold R² of an UNRESTRICTED OLS on the gene's real design (<code>card_context.py:15</code>). AUTHORED
ctx_collapse GENE REAL 0 100%	n_arm / n_fam for the gene “ how much curation BUNDLES same-seed mates onto it (1“5). AUTHORED
ctx_d_collin GENE SIGNED 1 84%	Within-design COLLINEARITY asymmetry, real minus decoy. The second design-match axis MH-167 tested; also bounded and not moving the gap. AUTHORED Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)
ctx_d_fam GENE REAL, SIGNED 99%	Design-WIDTH asymmetry between the real design and its matched decoy (real minus fake families). AUTHORED
ctx_dose_max GENE REAL 0 100%	dose maximum DERIVED
ctx_dose_med GENE REAL 0 100%	dose median DERIVED
ctx_frac_abund GENE FRACTION 0“1 100%	fraction unweighted abundance-sum reference DERIVED
ctx_gap_core GENE SIGNED 1 99%	gap core confounder block DERIVED
ctx_gap_d_dose GENE REAL, SIGNED 99%	Dose asymmetry between the real and decoy designs (median “0.43). AUTHORED
ctx_gap_deconv GENE SIGNED 1 99%	gap composition-adjusted (deconvolution block) DERIVED
ctx_gap_retention GENE REAL, SIGNED 95%	gap retention DERIVED
ctx_measurable GENE BOOLEAN 99%	Is the gene's <code>ctx_ceiling</code> above <code>CEILING_ROBUST</code> (0.02, not 0 “ MH-144)? AUTHORED

ctx_n_abund	count · unweighted abundance-sum reference	DERIVED
GENE COUNT 100%		
ctx_n_fam_abund	count · per seed family · unweighted abundance-sum reference	DERIVED
GENE COUNT 100%		
ctx_n_fam_multi	count · per seed family	DERIVED
GENE COUNT 100%		

IDENTITY_ · 5

supporting · read these when the core raises a question

5 columns · rung edge · coverage 96%~98% (median 96%)
2× fraction 0~1 · 2× real, signed · 1× boolean

identity_abs	identity as a fraction of the gene's total identity , clipped to [0,1] .	AUTHORED
EDGE FRACTION 0~1 96%		
identity_allocated	Family identity with phantom minor arms forced to 0 · the allocation actually used downstream.	AUTHORED
EDGE REAL, SIGNED 96%		
identity_coherence	$1 / \sum identity $ over the gene, clipped to (0,1] · how much the gene's identity shares CANCEL.	AUTHORED
EDGE FRACTION 0~1 96%		
identity_deconv	Shapley identity recomputed under the composition-adjusted design.	AUTHORED
EDGE REAL, SIGNED 98%		
identity_reliable	Is this edge's Shapley <code>identity</code> trustworthy · <code>identity_coherence</code> ≥ 0.5 AND identity ≥ 1?	AUTHORED
EDGE BOOLEAN 96%		

DOSE_ · 7

supporting · read these when the core raises a question

Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic.
7 columns · spans arm, edge · this prefix groups columns living on DIFFERENT units · coverage 55%~72% (median 72%) · 6 share the block description
3× count · 2× real ≥ 0 · 1× categorical · 1× constant

dose_class	class label	DERIVED
EDGE CATEGORICAL 72%	Defined on: edges present in the progression panel (arm expressed + state measured)	
dose_comp_retention	retention	DERIVED
ARM REAL ≥ 0 55%	Defined on: edges with a matched NAT leg (n~104 paired participants)	
dose_confounded	Is this edge's NAT's tumour dose a composition/proliferation artifact (either gated retention < 0.4)?	AUTHORED
ARM CONSTANT 72%		
dose_prolif_retention	retention	DERIVED
ARM REAL ≥ 0 55%	Defined on: edges with a matched NAT leg (n~104 paired participants)	
dose_rank_HLY	rank · in the GTEx-healthy state	DERIVED
EDGE COUNT 72%	Defined on: arms with a GTEX-HEALTHY leg (the thinnest state · many arms are multi-mapping-collapsed, MH-166)	

dose_rank_NAT	rank in matched normal (NAT) DERIVED
EDGE COUNT 72%	 Defined on: edges with a matched NAT leg (n~104 paired participants)
dose_rank_TUM	rank in tumour DERIVED
EDGE COUNT 72%	 Defined on: edges present in the progression panel (arm expressed + state measured)

D_RANK_ 4
supporting read these when the core raises a question

Change in the arm's WITHIN-GENE dose rank between two states.

4 columns in rung edge coverage 46%72% (median 72%) 3 share the block description
4x real, signed

d_rank_HLY_NAT	in the GTEx-healthy state in matched normal (NAT) DERIVED
EDGE REAL, SIGNED 72%	 Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)
d_rank_HLY_TUM	in the GTEx-healthy state in tumour DERIVED
EDGE REAL, SIGNED 72%	 Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)
d_rank_HLY_TUM_meas	d_rank_HLY_TUM recomputed over arms with a MEASURED healthy level only the collapse-safe version. Prefer it over the bare column. AUTHORED
EDGE REAL, SIGNED 46%	 Defined on: edges whose gene has a GTEx-MEASURED regulator NaN = GTEx cannot see it (abstention), NOT 0
d_rank_NAT_TUM	in matched normal (NAT) in tumour DERIVED
EDGE REAL, SIGNED 72%	 Defined on: edges with a matched NAT leg (n~104 paired participants)

GENE_ 6
supporting read these when the core raises a question

Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction.

6 columns in rung gene coverage 72%100% (median 99%) 3 share the block description
2x categorical 2x boolean 1x real 1x real, signed

gene_cancer_role	NOT the arm's role the GENE's cancer role (oncogene / tsg / oncogene/tsg / unknown), from gene_roles.load_gene_roles . AUTHORED
GENE CATEGORICAL 100%	
gene_dominated	Does one family carry more than 60% of this gene's tumour pressure budget (2,641 of 5,495)? AUTHORED
GENE BOOLEAN 97%	
gene_iqr	interquartile range DERIVED
GENE REAL 0 72%	 Defined on: genes with a NAT-TUM gene-level leg
gene_lfc_NAT_TUM	in matched normal (NAT) in tumour DERIVED
GENE REAL, SIGNED 72%	 Defined on: genes with a NAT-TUM gene-level leg
gene_net_repr	Is this gene NET-REPPRESSED in tumour by the 6b-RETIRED heuristic lane (1,376 of 5,634 edges)? AUTHORED
GENE BOOLEAN 100%	
gene_repr_class	class label DERIVED
GENE CATEGORICAL 100%	

AGO_ 2

supporting read these when the core raises a question

2 columns

rungrung

arm

ago_loading is rank-identical to the arm card's ago_dom one quantity, two names

coverage 95%100% (median 97%)

1x fraction 1 1x boolean

ago_loading

ARM

FRACTION 1

95%

AGO/RISC loading discount.

AUTHORED

ago_loading_measured

ARM

BOOLEAN

100%

Was ago_loading actually MEASURED for this edge (vs defaulted to 1.0). True for 3,820/5,648 edges (67.6%).

AUTHORED

Defined on: arms with a Manakov AGO-dominance measurement NOT loading(), which fabricates 1.0

PIP_ 2

supporting read these when the core raises a question

2 columns

rungrung

edge

coverage 100%100% (median 100%)

1x constant 1x fraction 1

pip_dense

EDGE

CONSTANT

100%

CONSTANT by construction the coupling readout is called with 1, so every edge has the same value.

AUTHORED

pip_discovery

EDGE

FRACTION 1

100%

Posterior inclusion probability. pip_dense comes from the pi=1 coupling readout; pip_discovery from the evidence-pi readout.

AUTHORED

RETENTION_ 3

reference context, provenance and rarely-quoted detail

3 columns

rungrung

edge

coverage 72%100% (median 100%)

1x real 0 1x boolean 1x real, signed

retention_beta

EDGE

REAL 0

100%

Fraction of an effect surviving covariate adjustment (adjusted / raw).

AUTHORED

retention_reliable

EDGE

BOOLEAN

100%

Is this edge's retention ratio trustworthy i.e. does it have a real denominator?

AUTHORED

retention_rho

EDGE

REAL, SIGNED

72%

deconv / core

AUTHORED

OOF_ 3

reference context, provenance and rarely-quoted detail

Out-of-fold predictive correlation, arm-level vs family-level, and their difference (oof_drho = arm minus family). Negative = the arm-resolved fit predicts repression better.

3 columns

rungrung

family

coverage 20%20% (median 20%)

2 share the block description

3x signed 1

oof_drho	Arm-level OOF rho MINUS family-level.	AUTHORED
FAMILY SIGNED $\hat{\rho}^{\prime 1} \hat{\rho}_{\epsilon} 1$		
20%		
oof_rho_arm	Spearman $\hat{\rho}$ per arm	DERIVED
FAMILY SIGNED $\hat{\rho}^{\prime 1} \hat{\rho}_{\epsilon} 1$	Defined on: multi-arm cells only ($n_{arm_in_cell} > 1$) $\hat{\rho}$ 20.3% of edges	
20%		
oof_rho_fam	Spearman $\hat{\rho}$ per seed family	DERIVED
FAMILY SIGNED $\hat{\rho}^{\prime 1} \hat{\rho}_{\epsilon} 1$	Defined on: multi-arm cells only ($n_{arm_in_cell} > 1$) $\hat{\rho}$ 20.3% of edges	
20%		

GRANK_ $\hat{\rho}$ 3

reference $\hat{\rho}$ context, provenance and rarely-quoted detail

The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from.

3

 columns $\hat{\rho}$ rung **arm** $\hat{\rho}$ coverage **63%** $\hat{\rho}$ **69%** (median 68%) $\hat{\rho}$ **3** share the block description

3x count

grank_HLY	in the GTEx-healthy state	DERIVED
ARM COUNT 63%	Defined on: arms with a GTEx-HEALTHY leg (the thinnest state $\hat{\rho}$ many arms are multi-mapping-collapsed, MH-166)	
grank_NAT	in matched normal (NAT)	DERIVED
ARM COUNT 68%	Defined on: edges with a matched NAT leg ($n \sim 104$ paired participants)	
grank_TUM	in tumour	DERIVED
ARM COUNT 69%	Defined on: edges present in the progression panel (arm expressed + state measured)	

COMPOSITION_ $\hat{\rho}$ 1

reference $\hat{\rho}$ context, provenance and rarely-quoted detail

1 columns $\hat{\rho}$ rung **edge** $\hat{\rho}$ coverage **100%** $\hat{\rho}$ **100%** (median 100%)

1x categorical

composition_class	Composition-adjustment verdict for this row (e.g. <code>composition_explained</code> when the effect does not survive the deconvolution block).	AUTHORED
EDGE CATEGORICAL 100%		

W_ $\hat{\rho}$ 1

reference $\hat{\rho}$ context, provenance and rarely-quoted detail

1 columns $\hat{\rho}$ rung **gene** $\hat{\rho}$ coverage **100%** $\hat{\rho}$ **100%** (median 100%)

1x real $\hat{\rho} \neq 0$

w_max	Maximum curated EVIDENCE weight over the gene's regulators. Gene-rung repeat.	AUTHORED
GENE REAL $\hat{\rho} \neq 0$ 100%		

OWN_ 1

reference “ context, provenance and rarely-quoted detail

1 columns rung arm coverage 88%88% (median 88%)

1x real 0

own_specific_frac

Fraction of the arm's variance that is target-specific. AUTHORED

ARM REAL 0 88%

DGLOBAL_ 3

reference “ context, provenance and rarely-quoted detail

Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step.

3 columns rung arm coverage 63%68% (median 63%) 3 share the block description

3x real, signed

dGlobal_HLY_NAT

in the GTEx-healthy state in matched normal (NAT) DERIVED

ARM REAL, SIGNED 63%

Defined on: arms with a GTEx-HEALTHY leg (the thinnest state “ many arms are multi-mapping-collapsed, MH-166)

dGlobal_HLY_TUM

in the GTEx-healthy state in tumour DERIVED

ARM REAL, SIGNED 63%

Defined on: arms with a GTEx-HEALTHY leg (the thinnest state “ many arms are multi-mapping-collapsed, MH-166)

dGlobal_NAT_TUM

in matched normal (NAT) in tumour DERIVED

ARM REAL, SIGNED 68%

Defined on: edges with a matched NAT leg (n~104 paired participants)

M_ 1

reference “ context, provenance and rarely-quoted detail

1 columns rung edge coverage 100%100% (median 100%)

1x real 0

m_nnlS

Bagged-NNLS weight. Exactly 0 in 31.7% of families “ this is what lets identity say a regulator contributes nothing, which beta structurally cannot. AUTHORED

EDGE REAL 0 100%

TERM_ 3

reference “ context, provenance and rarely-quoted detail

3 columns rung edge coverage 68%68% (median 68%)

2x signed 11 1x real, signed

term_ABUND

Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms. AUTHORED

EDGE SIGNED 11 68%

Defined on: edges present in the progression panel (arm expressed + state measured)

term_INTERACT

The INTERACTION term “ the part of the NAT’tumour change attributable to abundance and wiring moving TOGETHER. AUTHORED

EDGE SIGNED 11 68%

term_WIRING EDGE REAL, SIGNED68%	The WIRING term of the dose-vs-wiring decomposition: how much of the NAT’s tumour change comes from the arm's M-weight moving rather than its abundance. AUTHORED
PRIOR_ 1 reference “ context, provenance and rarely-quoted detail	
1 columns rung edge coverage 100%“100% (median 100%) 1x constant	
prior_pi EDGE CONSTANT100%	The Gibbs sampler's inclusion prior “ CONSTANT 0.05 across the card. AUTHORED
EDGE_ 1 reference “ context, provenance and rarely-quoted detail	
1 columns rung edge coverage 73%“73% (median 73%) 1x signed “1“1	
edge_rho_adj EDGE SIGNED “1“173%	An edge-rung quantity carried onto this row. AUTHORED Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)
MEAN_ 2 reference “ context, provenance and rarely-quoted detail	
2 columns rung arm coverage 90%“90% (median 90%) 2x real, signed	
mean_dGlobalRank ARM REAL, SIGNED90%	Mean change in the arm's GLOBAL abundance percentile across the paired patients. AUTHORED
mean_own_shift ARM REAL, SIGNED90%	A per-row mean of a lower-rung quantity; check agg_of for which rung it summarises. AUTHORED Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)
SURROGATE_ 2 reference “ context, provenance and rarely-quoted detail	
2 columns rung arm coverage 3%“3% (median 3%) 1x fraction 0“1 1x categorical	
surrogate_corr ARM FRACTION 0“13%	How well the surrogate tracks the arm where both are observed (0.54 “ 0.89). AUTHORED
surrogate_instrument ARM CATEGORICAL3%	Which SURROGATE stands in for a collapsed GTEx healthy leg (7 levels; defined on 3% of edges). AUTHORED

ESUB_ 8

reference “ context, provenance and rarely-quoted detail

PAM50 subtype-stratified coupling at the EDGE rung “ one fit per subtype plus the pooled fit and an adjusted q.

8 columns rung edge coverage 98%–100% (median 99%) 5 share the block description

6× signed 1 1 1× categorical 1× p-value

esub_best_sub EDGE CATEGORICAL 100%	the best DERIVED Defined on: edges with a per-PAM50 heterogeneity fit “ ăš DISTRIBUTIONAL (MH-165), not a per-edge subtype label
esub_q_adj EDGE P-VALUE 98%	BH-adjusted q for the best-subtype contrast “ AUTHORED
esub_rho_Basal EDGE SIGNED 1 1 99%	Spearman Ĩ DERIVED Defined on: edges with a per-PAM50 heterogeneity fit “ ăš DISTRIBUTIONAL (MH-165), not a per-edge subtype label
esub_rho_Her2 EDGE SIGNED 1 1 98%	Spearman Ĩ DERIVED Defined on: edges with a per-PAM50 heterogeneity fit “ ăš DISTRIBUTIONAL (MH-165), not a per-edge subtype label
esub_rho_LumA EDGE SIGNED 1 1 99%	Spearman Ĩ DERIVED Defined on: edges with a per-PAM50 heterogeneity fit “ ăš DISTRIBUTIONAL (MH-165), not a per-edge subtype label
esub_rho_LumB EDGE SIGNED 1 1 99%	Spearman Ĩ DERIVED Defined on: edges with a per-PAM50 heterogeneity fit “ ăš DISTRIBUTIONAL (MH-165), not a per-edge subtype label
esub_rho_best EDGE SIGNED 1 1 100%	Coupling in the arm's BEST PAM50 subtype. AUTHORED
esub_rho_pool EDGE SIGNED 1 1 100%	Coupling pooled across subtypes “ the unselected reference for esub_rho_best (median ăˆ0.0135, indistinguishable from a random-subtype draw at ăˆ0.0123). AUTHORED

REALIZATION_ 2

reference “ context, provenance and rarely-quoted detail

2 columns rung edge coverage 58%–69% (median 63%)

1× fraction 0 1 1× real, signed

realization_identity EDGE FRACTION 0 1 58%	Within-patient realization quantities “ the score, its identity allocation, and who owns it. AUTHORED Defined on: genes with a family-level realization fit
realization_score EDGE REAL, SIGNED 69%	ăˆcoupling_z_tum “ the calibrated tumour repression strength as a positive score, so bigger = more repressed. AUTHORED

KD_ 2

reference context, provenance and rarely-quoted detail

2 columns

rung

edge

coverage

85%85% (median 85%)

1x percent

1x real, signed

kd_affinity_pct

EDGE

PERCENT

85%

Is this gene among the arm's strongest predicted targets a WITHIN-ARM percentile, higher = stronger target. AUTHORED

kd_repression

EDGE

REAL, SIGNED

85%

scanMiR RBNS binding affinity. AUTHORED

WIRING_ 1

reference context, provenance and rarely-quoted detail

1 columns

rung

edge

coverage

68%68% (median 68%)

1x fraction

01

wiring_frac

EDGE

FRACTION 01

68%

Fraction of the acquisition score attributable to WIRING rather than abundance. AUTHORED

Defined on: edges present in the progression panel (arm expressed + state measured)

CELL_ 1

reference context, provenance and rarely-quoted detail

1 columns

rung

family

coverage

20%20% (median 20%)

1x boolean

cell_arms_resolvable

FAMILY

BOOLEAN

20%

Are the cell's arms separable AND does splitting them help out-of-fold. True for 31.7% of edges in multi-arm cells. AUTHORED

SD_ 1

reference context, provenance and rarely-quoted detail

1 columns

rung

edge

coverage

20%20% (median 20%)

1x real

0

sd_arm

EDGE

REAL 0

20%

Posterior SD of the arm-level I². AUTHORED

Z_ 1

reference context, provenance and rarely-quoted detail

1 columns

rung

edge

coverage

20%20% (median 20%)

1x real

0

z_arm

EDGE

REAL 0

20%

beta_arm / sd_arm from the WITHIN-CELL fit (arm_rung.py) the arm's z inside its (gene, seed_family) cell only, defined on the 20.3% of edges in multi-arm cells. AUTHORED

reference “ context, provenance and rarely-quoted detail

acquisition_score

Composite acquisition score across states.

AUTHORED

EDGE REAL, SIGNED 59%

Defined on: arms with a GTE_x-HEALTHY leg (the thinnest state $\hat{a} \in$ "many arms are multi-mapping-collapsed, MH-166)

reference " context, provenance and rarely-quoted detail

dShare M own

Mean over the 103 paired patients of this arm's change in M-WEIGHTED share of its gene's total regulator dose, tumour minus that patient's OWN NAT. AUTHORED

AUTHORED

EDGE	SIGNED $\hat{A}^{'1} \hat{A} \epsilon ! 1$	65%
------	--	-----

dShare_raw_own

As dShare_M_own but on RAW abundance share, with the model's M weights removed. AUTHORED

EDGE	SIGNED $\hat{A}^{-1} \in \mathbb{R}^{1 \times 1}$	65%
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reference “ context, provenance and rarely-quoted detail

heur_gene_nreg

Regulator count for this gene from the $\S 6b$ -RETIRED heuristic lane.

AUTHORED

GENE COUNT 100%

reference " context, provenance and rarely-quoted detail

family_rho_adj

Family-level properties carried onto this row (size, the arm's dose share of it, and its role). AUTHORED

AUTHORED

FAMILY SIGNED $\hat{A}'1\hat{A}\in|1$

69%

reference “ context, provenance and rarely-quoted detail

CPTAC protein channel.

20 columns Â· rung **edge** Â· per-edge; the gene card aggregates it Â· coverage **45%â€”79%** (median 65%) Â· **16**
 share the block description
 12x signed â€”1â€”1 Â· 4x real, signed Â· 2x constant Â· 2x categorical

cptac_prosp_comp_class_prot EDGE CATEGORICAL 59% AUTHORED	Composition class of this edge's PROTEIN coupling (cell_intrinsic / partial / composition_explained) in the prospective cohort. AUTHORED
cptac_prosp_n EDGE CONSTANT 79% AUTHORED	Patients in the prospective CPTAC cohort â€” CONSTANT 101 . AUTHORED
cptac_prosp_ret_prot EDGE REAL, SIGNED 59% DERIVED	CPTAC prospective cohort (nâ‰‰^101) Â· against protein Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_ret_rna EDGE REAL, SIGNED 61% DERIVED	CPTAC prospective cohort (nâ‰‰^101) Â· against mRNA Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_rho_disc EDGE SIGNED Âˆ'1Â€ 1 78% DERIVED	CPTAC prospective cohort (nâ‰‰^101) Â· Spearman Ĩ Â· against the discovery layer Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_rho_disc_raw EDGE SIGNED Âˆ'1Â€ 1 78% DERIVED	CPTAC prospective cohort (nâ‰‰^101) Â· Spearman Ĩ Â· against the discovery layer Â· UNADJUSTED â€” no confounder block Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_rho_prot EDGE SIGNED Âˆ'1Â€ 1 78% DERIVED	CPTAC prospective cohort (nâ‰‰^101) Â· Spearman Ĩ Â· against protein Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_rho_prot_raw EDGE SIGNED Âˆ'1Â€ 1 79% AUTHORED	UNADJUSTED protein coupling, prospective. AUTHORED
cptac_prosp_rho_rna EDGE SIGNED Âˆ'1Â€ 1 79% DERIVED	CPTAC prospective cohort (nâ‰‰^101) Â· Spearman Ĩ Â· against mRNA Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_rho_rna_raw EDGE SIGNED Âˆ'1Â€ 1 79% DERIVED	CPTAC prospective cohort (nâ‰‰^101) Â· Spearman Ĩ Â· against mRNA Â· UNADJUSTED â€” no confounder block Defined on: genes/edges present in the CPTAC cohort
cptac_t105_comp_class_prot EDGE CATEGORICAL 45% DERIVED	CPTAC TCGA-overlap cohort (n=105) Â· class label Â· against protein Defined on: genes/edges present in the CPTAC cohort
cptac_t105_n EDGE CONSTANT 70% AUTHORED	Patients in the TCGA-overlap CPTAC cohort â€” CONSTANT 105 . See the prospective twin. AUTHORED
cptac_t105_ret_prot EDGE REAL, SIGNED 45% DERIVED	CPTAC TCGA-overlap cohort (n=105) Â· against protein Defined on: genes/edges present in the CPTAC cohort
cptac_t105_ret_rna EDGE REAL, SIGNED 49% DERIVED	CPTAC TCGA-overlap cohort (n=105) Â· against mRNA Defined on: genes/edges present in the CPTAC cohort
cptac_t105_rho_disc EDGE SIGNED Âˆ'1Â€ 1 64% DERIVED	CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ĩ Â· against the discovery layer Defined on: genes/edges present in the CPTAC cohort
cptac_t105_rho_disc_raw EDGE SIGNED Âˆ'1Â€ 1 65% DERIVED	CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ĩ Â· against the discovery layer Â· UNADJUSTED â€” no confounder block Defined on: genes/edges present in the CPTAC cohort
cptac_t105_rho_prot EDGE SIGNED Âˆ'1Â€ 1 64% DERIVED	CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ĩ Â· against protein Defined on: genes/edges present in the CPTAC cohort
cptac_t105_rho_prot_raw EDGE SIGNED Âˆ'1Â€ 1 65% DERIVED	CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ĩ Â· against protein Â· UNADJUSTED â€” no confounder block Defined on: genes/edges present in the CPTAC cohort

EDGE SIGNED $\hat{A}'_1 \in \mathbb{R}^{1 \times 1}$ 64%

Defined on: genes/edges present in the CPTAC cohort

EDGE SIGNED $\hat{A}^{\prime} \in \mathbb{R}^{n \times n}$ 65%

Defined on: genes/edges present in the CPTAC cohort

reference “ context, provenance and rarely-quoted detail

```
1 columns  Â·  rung edge  Â·  coverage 61%â€”61% (median 61%)
1x boolean
```

EDGE BOOLEAN 61%

DISCORDANT 23.9% of the 3,433 edges carrying both. AUTHORED

reference “ context, provenance and rarely-quoted detail

```
1 columns  Â·  rung edge  Â·  coverage 58%â€”58% (median 58%)
1x boolean
```

EDGE BOOLEAN 58%

Defined on: genes with a family-level realization fit

realization/gene_card.tsv

gene

One row per the gene's total incoming regulation. One row per gene, so nothing here is checkable by invariance – `agg_of` carries the meaning instead.

135 columns 1,549 rows 29 blocks median coverage 83%

Contents – 29 blocks, most important first

1. n_ 7 – CORE

2. (bare) 2 – CORE

3. comp_ 16 – CORE

4. ctx_ 16 – CORE

5. top_ 8 – SUPPORTING

6. realized_ 5 – SUPPORTING

7. w_ 1 – SUPPORTING

8. lit_ 7 – SUPPORTING

9. cptac_ 24 – REFERENCE

10. frac_ 3 – REFERENCE

11. dominant_ 4 – REFERENCE

12. static_ 1 – REFERENCE

13. total_ 1 – REFERENCE

14. owner_ 1 – REFERENCE
15. median_ 1 – REFERENCE

16. armres_ 7 – REFERENCE

17. identity_ 1 – REFERENCE

18. heur_ 5 – REFERENCE

19. greal_ 10 – REFERENCE

20. acquired_vs_ 2 – REFERENCE

21. pressure_ 1 – REFERENCE

22. realization_ 1 – REFERENCE

23. tcga_ 4 – REFERENCE

24. disc_ 2 – REFERENCE

25. max_ 1 – REFERENCE

26. regulatory_ 1 – REFERENCE

27. mean_ 1 – REFERENCE

28. concentration_ 1 – REFERENCE

29. fam_ 1 – REFERENCE

N_ – 7

core – the columns findings rest on

7 columns – rung gene – coverage 91%–100% (median 100%)
7x count

n_arms

GENE COUNT AGG ARM 100%

Arms in the LEARNED MODEL's design – `X.shape[1]` from `assemble_gene`, carried via `gene_atlas`. **This is the width the Gibbs actually fit** (median 2, max 90). AUTHORED

n_cell_intrinsic

GENE COUNT 100%

FAMILIES whose coupling SURVIVES the composition block (retention ≥ 0.7). AUTHORED

n_composition_explained

GENE COUNT 100%

FAMILIES whose coupling does NOT survive the composition block (retention < 0.4). AUTHORED

n_fam

GENE COUNT AGG FAMILY
100%

How many seed families regulate this gene. AUTHORED

n_hallmark_sets

GENE COUNT 91%

How many MSigDB Hallmark programs contain this gene (1–10, median 2). Membership only – it says nothing about regulation. AUTHORED

n_identified

GENE COUNT 100%

Families reaching $|z| > 2$ on the UNCALIBRATED SD. **Zero for 691 genes (44.6%)** – meaning the model cannot distinguish ANY of that gene's families from zero. AUTHORED

n_pip_disc_gt50 GENE COUNT 100%	Count of the gene's families with <code>pip_discovery > 0.5</code> under the evidence-İreadout. AUTHORED
(BARE) Å 2 core â€ the columns findings rest on	
2 columns Å spans gene, key â€ this prefix groups columns living on DIFFERENT units Å coverage 100%â€100% (median 100%) 1x fraction 0â€1 Å 1x identifier	
concentration GENE FRACTION 0â€1 100%	<code>beta.max() / beta.sum()</code> â€ the TOP FAMILY's share of the gene's total İ²; median 0.954, i.e. most genes are dominated by one family. AUTHORED
gene KEY IDENTIFIER 100%	KEY â€ HGNC gene symbol. AUTHORED
COMP_ Å 16 core â€ the columns findings rest on	
Cell-composition drivers of this gene's coupling (<code>compartment_*</code>): which deconvolved fraction explains the correlation, its retention under adjustment, and its share. 16 columns Å rung gene Å coverage 59%â€90% (median 73%) Å 7 share the block description 6x categorical Å 6x signed â^'1â€1 Å 3x real, signed Å 1x real â%¥ 0	
comp_cptac_prot_driver GENE CATEGORICAL 59%	CPTAC Å against protein DERIVED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_driver_ret GENE REAL, SIGNED 59%	1 â^' comp_cptac_prot_driver_share, exactly. Median +0.398 raw, +0.469 gated to İ_raw â%¥ 0.10 â€ quote the gated value. See the mRNA twin. AUTHORED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_driver_share GENE REAL, SIGNED 59%	Protein-layer twin of <code>comp_tcga_mrna_driver_share</code>. Same exact complementarity with <code>_driver_ret</code> (8.9e-16 on 918 rows). AUTHORED
comp_cptac_prot_rho_raw GENE SIGNED Å^'1â€1 73%	Unadjusted aggregate PROTEIN coupling â€ denominator of <code>comp_cptac_prot_driver_ret</code>. AUTHORED
comp_cptac_prot_top_budget GENE CATEGORICAL 73%	CPTAC Å against protein Å the top-ranked Å pressure budget DERIVED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_budget_r GENE SIGNED Å^'1â€1 73%	CPTAC Å against protein Å the top-ranked Å pressure budget Å correlation DERIVED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_target GENE CATEGORICAL 73%	CPTAC Å against protein Å the top-ranked DERIVED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_target_r GENE SIGNED Å^'1â€1 73%	CPTAC Å against protein Å the top-ranked Å correlation DERIVED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)

comp_tcga_mrna_driver GENE CATEGORICAL 66%	WHICH compartment drives the gene's mRNA coupling â€” the one whose removal moves it most (CAFs 279 Â· purity 174 Â· prolif 148 Â· Endothelial 122 Â· Myeloid 109). AUTHORED
comp_tcga_mrna_driver_ret GENE REAL, SIGNED 66%	1 ^ ' comp_tcga_mrna_driver_share , exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. AUTHORED
comp_tcga_mrna_driver_share GENE REAL Â% 0 66%	Share of the gene's mRNA coupling attributable to the single compartment that most changes it â€” axiom 8's composition GATE , the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. AUTHORED
comp_tcga_mrna_rho_raw GENE SIGNED Â^ ' 1 Â€ 1 90%	The gene's UNADJUSTED aggregate mRNA coupling â€” the DENOMINATOR of comp_tcga_mrna_driver_ret . AUTHORED
comp_tcga_mrna_top_budget GENE CATEGORICAL 90%	The compartment the gene's miRNA BUDGET loads on most â€” a property of the REGULATORS. AUTHORED
comp_tcga_mrna_top_budget_r GENE SIGNED Â^ ' 1 Â€ 1 90%	TCGA Â· the top-ranked Â· pressure budget Â· correlation DERIVED Defined on: genes/families with a TCGA-mRNA compartment driver â€” âŒš MH-201: these axes predict the SIGN of Î²'s gain
comp_tcga_mrna_top_target GENE CATEGORICAL 90%	The compartment the TARGET GENE's expression loads on most. AUTHORED
comp_tcga_mrna_top_target_r GENE SIGNED Â^ ' 1 Â€ 1 90%	TCGA Â· the top-ranked Â· correlation DERIVED Defined on: genes/families with a TCGA-mRNA compartment driver â€” âŒš MH-201: these axes predict the SIGN of Î²'s gain

CTX_ Â· 16
core â€” the columns findings rest on

Gene-design context from the decoy/atlas layer (learned/analyses/gene_atlas.py via card_context): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

16 columns Â· rung **gene** Â· SUBSET of the edge block, identical rungs Â· coverage **45%â€”100%** (median 90%) Â· **11** share the block description

4x signed Â^ ' 1 Â€ | 1 Â· 3x real Â% 0 Â· 3x real, signed Â· 3x count Â· 1x categorical Â· 1x fraction 0â€”1 Â· 1x boolean

ctx_apriori_class GENE CATEGORICAL 100%	A-priori competence class from the gene atlas (A_COMPETENT / B_PARTIAL / C_WEAK / D_UNDEFINED). AUTHORED
ctx_ceiling GENE SIGNED Â^ ' 1 Â€ 1 98%	The UPPER BOUND on what ANY estimator can achieve for this gene â€” the 5-fold out-of-fold RÂ² of an UNRESTRICTED OLS on the gene's real design (card_context.py:15). AUTHORED
ctx_collapse GENE REAL Â% 0 90%	n_arm / n_fam for the gene â€” how much curation BUNDLES same-seed mates onto it (1â€”5). AUTHORED
ctx_d_collin GENE SIGNED Â^ ' 1 Â€ 1 45%	change DERIVED Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)
ctx_d_fam GENE REAL, SIGNED 87%	change Â· per seed family DERIVED
ctx_dose_max GENE REAL Â% 0 90%	dose Â· maximum DERIVED

ctx_dose_med GENE REAL 0 90%	dose · median DERIVED
ctx_frac_abund GENE FRACTION 0 1 90%	fraction · unweighted abundance-sum reference DERIVED
ctx_gap_core GENE SIGNED 1 87%	Real-minus-decoy coupling gap for this gene. Pooled it is ~-0.012, but the per-gene IQR STRADDLES ZERO – this is a set-level distributional shift, not a per-gene effect. Do not rank genes by it. AUTHORED
ctx_gap_d_dose GENE REAL, SIGNED 87%	gap · change · dose DERIVED
ctx_gap_deconv GENE SIGNED 1 87%	gap · composition-adjusted (deconvolution block) DERIVED
ctx_gap_retention GENE REAL, SIGNED 83%	gap · retention DERIVED
ctx_measurable GENE BOOLEAN 98%	Is the gene's ctx_ceiling above CEILING_ROBUST (0.02, not 0 – MH-144)? AUTHORED
ctx_n_abund GENE COUNT 90%	count · unweighted abundance-sum reference DERIVED
ctx_n_fam_abund GENE COUNT 90%	count · per seed family · unweighted abundance-sum reference DERIVED
ctx_n_fam_multi GENE COUNT 90%	count · per seed family DERIVED

TOP · 8
supporting – read these when the core raises a question

The gene's leading regulator and its value, by magnitude or by identity.

8 columns · rung gene · coverage 78–100% (median 89%) · 3 share the block description
3× identifier · 2× real 0 · 1× fraction 0–1 · 1× signed 1 · 1× count

top_beta GENE REAL 0 AGG FAMILY 100%	$\hat{\beta}^2$ DERIVED
top_beta_frac GENE FRACTION 0 1 AGG FAMILY 100%	$\hat{\beta}^2$ · fraction DERIVED
top_family_identity GENE IDENTIFIER AGG FAMILY 87%	The family Shapley/LMG credits with the gene's regulation. NaN for 197 genes (12.7%) where NNLS zeroed everything – an honest 'undefined', which the magnitude answer cannot express. AUTHORED
top_family_magnitude GENE IDENTIFIER AGG FAMILY 100%	The family with the largest beta. AUTHORED

top_gaining_arms	arms	DERIVED
GENE	IDENTIFIER	91%
top_identity	The largest identity share among the gene's families. AUTHORED	
GENE	REAL $\in [0, 1]$	
AGG FAMILY	87%	
top_identity_gated	top_identity recomputed over identity_reliable edges only $\hat{=}$ max 1.0000, 0 genes above 1 (vs +740.007 ungated). Defined on 1,203 genes. AUTHORED	
GENE	SIGNED $\in \{-1, 1\}$	78%
top_identity_n_reliable	How many of the gene's families survive the identity gate $\hat{=}$ the DENOMINATOR behind top_identity_gated. Median 2. AUTHORED	
GENE	COUNT	78%

REALIZED_ $\in [0, 1]$
supporting $\hat{=}$ read these when the core raises a question

Realized (as opposed to potential) coupling for the gene $\hat{=}$ the raw and adjusted rho, its retention, and how many regulators it was computed over.

5 columns $\hat{=}$ **run** **gene** $\hat{=}$ **coverage** **63%** $\hat{=}$ **82%** (median 81%) $\hat{=}$ **3** share the block description

2x signed $\in \{-1, 1\}$ $\hat{=}$ 1x boolean $\hat{=}$ 1x count $\hat{=}$ 1x real, signed

realized_composition_explained	Is the gene's REALIZED coupling composition-explained (retention < 0.4)? AUTHORED	
GENE	BOOLEAN	82%
realized_n_reg	count	DERIVED
GENE	COUNT	82%
realized_retention	Realized-coupling retention, $\hat{I}_{\text{adjusted}} / \hat{I}_{\text{raw}}$. AUTHORED	
GENE	REAL, SIGNED	63%
realized_rho_adj	Spearman $\hat{I}$ $\hat{=}$ composition-adjusted DERIVED	
GENE	SIGNED $\in \{-1, 1\}$	81%
realized_rho_raw	Spearman $\hat{I}$ $\hat{=}$ UNADJUSTED $\hat{=}$ no confounder block DERIVED	
GENE	SIGNED $\in \{-1, 1\}$	81%

W_ $\in [0, 1]$
supporting $\hat{=}$ read these when the core raises a question

1 columns $\hat{=}$ **run** **gene** $\hat{=}$ **coverage** **100%** $\hat{=}$ **100%** (median 100%)

1x real $\in [0, 1]$

w_max	The MAXIMUM curated evidence weight w over the gene's regulators (gene_atlas : nanmax(w)), where w comes from the same PMID-deduped ledger that feeds lit_ and fame_ . AUTHORED	
GENE	REAL $\in [0, 1]$	100%

LIT_ 7

supporting “ read these when the core raises a question

7 columns · rung gene · coverage 21%”21% (median 21%) 4× count · 2× boolean · 1× identifier		
lit_agrees_identity	Does <code>identity</code> name the same family as the most-published one for this gene “ True on 100 of 323 (31%) . AUTHORED	
lit_agrees_magnitude	Does the model's largest- $\hat{I}^2$ family match the most-published one (96 of 329 genes, 29%)? AUTHORED	
lit_family	The most-published seed family for this gene by distinct low-throughput functional PMIDs. AUTHORED	
lit_margin	Gap between the top and second literature family, in PMIDs “ median 1.00 , i.e. the literature's 'winner' is usually decided by a SINGLE paper. “ gate on this before reading <code>lit_agrees_*</code> : a 1-PMID margin is a coin flip dressed as ground truth. AUTHORED	<i>Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)</i>
lit_n_lit_families	“” A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH “ read the name with suspicion. <code>eval/lit_ground_truth.py</code> defines the 'canonical' family of a gene as the argmax of DISTINCT PubMed IDs carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded). AUTHORED	
lit_n_pmids	Distinct low-throughput functional PMIDs behind <code>lit_family</code> . Median 3 “ the 'ground truth' rests on a handful of papers per gene. AUTHORED	<i>Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)</i>
lit_tier	Literature-evidence tier for the gene (levels 2, 3 and 4; 4 is the most populated at 132 genes). AUTHORED	

CPTAC_ 24

reference “ context, provenance and rarely-quoted detail

CPTAC protein channel. 24 columns · rung gene · AGGREGATE of the edge block via an <code>agg_/abund_ infix</code> “ NOT derivable (MH-262) · coverage 42%”89% (median 66%) · 18 share the block description 18× signed “1”1 · 2× boolean · 2× real, signed · 2× count		
cptac_prosp_abund_rho_disc	CPTAC prospective cohort ($n \approx 101$) · unweighted abundance-sum reference · Spearman $\hat{I}$ · against the discovery layer DERIVED	<i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_prosp_abund_rho_prot	CPTAC prospective cohort ($n \approx 101$) · unweighted abundance-sum reference · Spearman $\hat{I}$ · against protein DERIVED	<i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_prosp_abund_rho_rna	CPTAC prospective cohort ($n \approx 101$) · unweighted abundance-sum reference · Spearman $\hat{I}$ · against mRNA DERIVED	<i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_prosp_agg_beats_abund_prot	Does the learned $\hat{I}^2$ -weighted aggregate couple more negatively to PROTEIN than a plain abundance sum, in the CPTAC prospective cohort? AUTHORED	

cptac_prosp_agg_ret_prot GENE REAL, SIGNED 59%	rho_prot(composition-adjusted) / rho_prot_raw “ the ADJUSTMENT sense of retention, on the prospective cohort. “... The denominator is ALREADY GATED at RHO_GATE in cptac_card.py:322 , so values outside [0,1] are NOT the vanishing-denominator artifact of axiom 5 (verified: 0 rows with raw < 0.02). AUTHORED
cptac_prosp_agg_rho_disc GENE SIGNED $\hat{A}^T \hat{A} \in \{1\}$ AGG ARM 72%	CPTAC prospective cohort ($n \propto 101$) $\hat{A} \cdot \hat{I}^2$-weighted aggregate $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against the discovery layer DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_agg_rho_disc_raw GENE SIGNED $\hat{A}^T \hat{A} \in \{1\}$ 72%	CPTAC prospective cohort ($n \propto 101$) $\hat{A} \cdot \hat{I}^2$-weighted aggregate $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against the discovery layer $\hat{A} \cdot \text{UNADJUSTED}$ “ no confounder block DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_agg_rho_prot GENE SIGNED $\hat{A}^T \hat{A} \in \{1\}$ AGG ARM 73%	CPTAC prospective cohort ($n \propto 101$) $\hat{A} \cdot \hat{I}^2$-weighted aggregate $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against protein DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_agg_rho_prot_raw GENE SIGNED $\hat{A}^T \hat{A} \in \{1\}$ 73%	CPTAC prospective cohort ($n \propto 101$) $\hat{A} \cdot \hat{I}^2$-weighted aggregate $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against protein $\hat{A} \cdot \text{UNADJUSTED}$ “ no confounder block DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_agg_rho_rna GENE SIGNED $\hat{A}^T \hat{A} \in \{1\}$ AGG ARM 74%	CPTAC prospective cohort ($n \propto 101$) $\hat{A} \cdot \hat{I}^2$-weighted aggregate $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against mRNA DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_agg_rho_rna_raw GENE SIGNED $\hat{A}^T \hat{A} \in \{1\}$ 74%	CPTAC prospective cohort ($n \propto 101$) $\hat{A} \cdot \hat{I}^2$-weighted aggregate $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against mRNA $\hat{A} \cdot \text{UNADJUSTED}$ “ no confounder block DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_n_arms GENE COUNT 89%	Arms contributing to this gene's prospective-cohort aggregate (median 2, max 61). AUTHORED
cptac_t105_abund_rho_disc GENE SIGNED $\hat{A}^T \hat{A} \in \{1\}$ 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A} \cdot$ unweighted abundance-sum reference $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against the discovery layer DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_t105_abund_rho_prot GENE SIGNED $\hat{A}^T \hat{A} \in \{1\}$ 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A} \cdot$ unweighted abundance-sum reference $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against protein DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_t105_abund_rho_rna GENE SIGNED $\hat{A}^T \hat{A} \in \{1\}$ 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A} \cdot$ unweighted abundance-sum reference $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against mRNA DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_t105_agg_beats_abund_prot GENE BOOLEAN 60%	As the prospective twin, on the TCGA-105 overlap cohort. AUTHORED
cptac_t105_agg_ret_prot GENE REAL, SIGNED 42%	As the prospective twin, TCGA-105. AUTHORED
cptac_t105_agg_rho_disc GENE SIGNED $\hat{A}^T \hat{A} \in \{1\}$ AGG ARM 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A} \cdot \hat{I}^2$-weighted aggregate $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against the discovery layer DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_t105_agg_rho_disc_raw GENE SIGNED $\hat{A}^T \hat{A} \in \{1\}$ 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A} \cdot \hat{I}^2$-weighted aggregate $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against the discovery layer $\hat{A} \cdot \text{UNADJUSTED}$ “ no confounder block DERIVED Defined on: genes/edges present in the CPTAC cohort

cptac_t105_agg_rho_prot GENE SIGNED $\hat{A}^{1\hat{A}} 1$ 60% AGG ARM 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\hat{I}^2$-weighted aggregate $\hat{A}$ Spearman $\hat{I}$ $\hat{A}$ against protein DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_t105_agg_rho_prot_raw GENE SIGNED $\hat{A}^{1\hat{A}} 1$ 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\hat{I}^2$-weighted aggregate $\hat{A}$ Spearman $\hat{I}$ $\hat{A}$ against protein $\hat{A}$ UNADJUSTED $\hat{A}$ no confounder block DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_t105_agg_rho_rna GENE SIGNED $\hat{A}^{1\hat{A}} 1$ 60% AGG ARM 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\hat{I}^2$-weighted aggregate $\hat{A}$ Spearman $\hat{I}$ $\hat{A}$ against mRNA DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_t105_agg_rho_rna_raw GENE SIGNED $\hat{A}^{1\hat{A}} 1$ 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\hat{I}^2$-weighted aggregate $\hat{A}$ Spearman $\hat{I}$ $\hat{A}$ against mRNA $\hat{A}$ UNADJUSTED $\hat{A}$ no confounder block DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_t105_n_arms GENE COUNT 85%	Arms behind the TCGA-105 aggregate (median 2, max 57). See the prospective twin. AUTHORED Defined on: genes/edges present in the CPTAC cohort

FRAC_ $\hat{A}$ 3

reference $\hat{A}$ context, provenance and rarely-quoted detail

3 columns $\hat{A}$ rung **gene** $\hat{A}$ coverage **60% $\hat{A}$ 100%** (median 82%)
3 $\times$ fraction $0\hat{A}$ 1

frac_edges_realized GENE FRACTION $0\hat{A}$1 82%	Fraction of the gene's scored edges that realize a coupling below the threshold. AUTHORED
frac_gain_true_acquired GENE FRACTION $0\hat{A}$1 60%	Fraction of the gene's acquired-repression calls that survive the healthy-leg triage $\hat{A}$ i.e. are not collapse_blind artifacts. AUTHORED
frac_identified GENE FRACTION $0\hat{A}$1 100%	Fraction of this gene's families reaching $z > 2$ on the UNCALIBRATED SD. AUTHORED

DOMINANT_ $\hat{A}$ 4

reference $\hat{A}$ context, provenance and rarely-quoted detail

The top-ranked regulator of this gene in one state (HLY / NAT / TUM), by GLOBAL dose rank. The inputs to **regulatory_handoff**.
4 columns $\hat{A}$ rung **gene** $\hat{A}$ coverage **77% $\hat{A}$ 83%** (median 82%) $\hat{A}$ 3 share the block description
3 $\times$ identifier $\hat{A}$ 1 $\times$ categorical

dominant_HLY GENE IDENTIFIER 77%	in the GTEx-healthy state DERIVED
dominant_NAT GENE IDENTIFIER 81%	in matched normal (NAT) DERIVED
dominant_TUM GENE IDENTIFIER 82%	The arm carrying the largest share of the gene's tumour pressure budget (miR-21-5p on 56 genes, miR-200c-3p next). AUTHORED
dominant_edge_shift_class GENE CATEGORICAL 83%	per edge $\hat{A}$ shift $\hat{A}$ class label DERIVED

STATIC_ 1

reference " context, provenance and rarely-quoted detail

1 columns rung gene coverage 87%87% (median 87%)

1x identifier

static_owner_family

GENE

IDENTIFIER

87%

The static (dose-based) owner of the gene, for comparison against the realization owner. AUTHORED

| Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)

TOTAL_ 1

reference " context, provenance and rarely-quoted detail

1 columns rung gene coverage 100%100% (median 100%)

1x real 0

total_pressure

GENE

REAL 0

100%

Sum over the gene's regulators. AUTHORED

OWNER_ 1

reference " context, provenance and rarely-quoted detail

1 columns rung gene coverage 41%41% (median 41%)

1x boolean

owner_agrees

GENE

BOOLEAN

41%

Whether the realization owner and the static owner agree. Defined only where both exist. AUTHORED

| Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)

MEDIAN_ 1

reference " context, provenance and rarely-quoted detail

1 columns rung gene coverage 100%100% (median 100%)

1x real 0

median_retention

GENE

REAL 0

100%

A per-row median of a lower-rung quantity; check agg_of. AUTHORED

ARMRES_ 7

reference " context, provenance and rarely-quoted detail

IS THIS GENE MODELABLE AT ARM RESOLUTION " the gene-level roll-up of arm identifiability (card_ladders.gene_arm_resolution, 2026-08-19).

7 columns rung gene coverage 18%18% (median 18%) 4 share the block description

2x signed 1 2x count 1x categorical 1x fraction 1 1x real 0

armres_best_drho

GENE

SIGNED 11

18%

the best DERIVED

armres_class	arm_modelable (every multi-arm cell resolvable, 68 genes) partial (37) family_only (no cell resolvable, 168). Absent = the gene has no multi-arm cell at all, so the question is undefined. AUTHORED
armres_frac_resolvable	Fraction of this gene's multi-arm cells where the arms are separable. AUTHORED
armres_med_drho	Median oof_drho over the gene's multi-arm cells â€” negative means arm resolution predicts repression better out of fold. AUTHORED
armres_med_sep_z	median â€¢ z-score DERIVED
armres_n_arms_split	count â€¢ arms DERIVED
armres_n_multi_cells	count DERIVED

IDENTITY_ **â€¢ 1**

reference **â€”** context, provenance and rarely-quoted detail

1 columns â€¢ rung gene â€¢ coverage 87%â€”87% (median 87%) 1x boolean

identity_eq_magnitude	Do the identity and magnitude answers name the SAME top family. Exactly 100% agreement at n_fam =1 by construction; 76% at n_fam >=2. Read it only on multi-family genes. AUTHORED
GENE BOOLEAN AGG FAMILY	
87%	

HEUR_ **â€¢ 5**

reference **â€”** context, provenance and rarely-quoted detail

5 columns â€¢ rung gene â€¢ coverage 89%â€”91% (median 91%) 2x signed â€”1â€”1 â€¢ 1x count â€¢ 1x boolean â€¢ 1x categorical
--

heur_delta_tumor_nat	Tumour-minus-NAT change in the heuristic pressure correlation. AUTHORED
GENE SIGNED â€”1â€”1	89%
heur_n_regulators	â€”â€” A DIFFERENT PIPELINE'S COUNT â€” not the model's. It is <code>edges.groupby('gene')['miRNA'].nunique()</code> inside <code>analyses/misc/mirna_comovement.py</code> , i.e. AUTHORED
GENE COUNT	91%
heur_net_repressed_tumor	Is the gene net-repressed in tumour, per the RETIRED heuristic lane. True for 282 of 1,409 genes. AUTHORED
GENE BOOLEAN	91%
heur_repression_class	Cross-state repression class from the RETIRED heuristic lane (<code>never_repressed</code> / <code>lost_repression</code> / <code>gained_repression</code> / <code>constitutive_repressed</code>). 867 of 1,409 genes are <code>never_repressed</code> . AUTHORED
GENE CATEGORICAL	91%
heur_rho_pressure_tumor	Spearman of the gene's HEURISTIC aggregate pressure against its mRNA in tumour. AUTHORED
GENE SIGNED â€”1â€”1	91%

GREAL_ 10

reference “ context, provenance and rarely-quoted detail

Realization ladder for the GENE “ the mirror of `real_` : how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). æ... Cross-rung control: the gene-side totals reproduce the arm-side exactly.

10 columns rung **gene** coverage **29%æ83%** (median 83%) 8 share the block description

7x count 2x signed 1 fraction 0æ1

greal_best_coupling GENE SIGNED 11 83%	the best DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n3
greal_frac_c10 GENE FRACTION 01 29%	Gene-rung twin of <code>real_frac_c10</code> “ defined on 446 genes against <code>greal_n_c10</code>'s 1,289; gated-out genes have a median <code>greal_n_scored</code> of 1. Exact on 100%. AUTHORED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n3
greal_med_coupling GENE SIGNED 11 83%	median DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n3
greal_n_c05 GENE COUNT 83%	count at the 0.05 threshold DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n3
greal_n_c10 GENE COUNT 83%	count at the 0.10 threshold DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n3
greal_n_c15 GENE COUNT 83%	count at the 0.15 threshold DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n3
greal_n_c20 GENE COUNT 83%	count at the 0.20 threshold DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n3
greal_n_c30 GENE COUNT 83%	count at the 0.30 threshold DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n3
greal_n_scored GENE COUNT 83%	Genes/edges scored for realized coupling (median 2, max 59) “ the DENOMINATOR of every <code>greal_frac_*</code>. æ... The <code>greal_n_c05</code> æ <code>n_c30</code> ladder is properly nested: 0 monotonicity violations and 0 rows where a threshold count exceeds <code>n_scored</code> (verified). AUTHORED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n3
greal_n_sig GENE COUNT 83%	count DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) “ the GENE-side ladder; rates gated n3

ACQUIRED_VS_ 2

reference “ context, provenance and rarely-quoted detail

Acquisition of regulatory dose relative to a healthy reference (GTEx or NAT).

2 columns rung **gene** coverage **91%æ91%** (median 91%) 2 share the block description

2x real, signed

GTE_x DERIVED

GENE REAL, SIGNED 91%

in matched normal (NAT) DERIVED

GENE REAL, SIGNED 91%

reference “ context, provenance and rarely-quoted detail

```
1 columns  Â·  rung gene  Â·  coverage 91%â€”91% (median 91%)
1x real  â‰¥ 0
```

Aggregate incoming pressure on the gene in the named state. AUTHORED

GENE REAL 0 91%

reference “ context, provenance and rarely-quoted detail

```
1 columns  Â·  rung gene  Â·  coverage 42%â€”42% (median 42%)
1x identifier
```

Within-patient realization quantities are the score, its identity allocation, and who owns it. AUTHORED

Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)

reference “ context, provenance and rarely-quoted detail

The TCGA mRNA leg carried beside the CPTAC columns so the two are comparable on the same gene rows.

4 columns: `Â·` `rung` `gene` `Â·` `coverage` `89%` `89%` (median 89%) `Â·` `3` share the block description
 3x signed `â`1â€1` `Â·` 1x count

unweighted abundance-sum reference $\hat{A}$ · Spearman ρ $\hat{A}$ · against mRNA DERIVED

GENE SIGNED $\hat{\alpha}'_1 \in [-1, 1]$

AGG ARM 89%

Gene-level $\hat{\tau}^2$ -weighted aggregate coupling to mRNA in TCGA (median $\hat{\alpha}^{\text{0.05}}$) against tcga_abund_rho_rna's $\hat{\alpha}^{\text{0.02}}$ the abundance-sum reference. AUTHORED

GENE SIGNED $\hat{\alpha}'_1 \in \mathbb{R}^1$

AGG ARM 89%

Î²-weighted aggregate $\hat{\Lambda}$ · Spearman $\tilde{\Lambda}$ · against mRNA $\hat{\Lambda}$ · UNADJUSTED “no confounder block” DERIVED

GENE SIGNED $\hat{\alpha}'_1 \in \mathbb{R}^1$ 89%

count $\hat{A}$ · arms DERIVED

GENE COUNT 89%

DISC_ 1 2

reference " context, provenance and rarely-quoted detail

2 columns 1 rung gene 1 coverage 3%100% (median 51%)
1x categorical 1x count

disc_gold_families

Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). AUTHORED

GENE CATEGORICAL 3%

disc_n_gold_edges

How many of the 157 discovery-queue edges name this gene (max 4). AUTHORED

GENE COUNT 100%

MAX_ 1 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung gene 1 coverage 100%100% (median 100%)
1x fraction 01

max_beta_frac_sd

Maximum over the row's constituent units. AUTHORED

GENE FRACTION 01 100%

REGULATORY_ 1 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung gene 1 coverage 100%100% (median 100%)
1x boolean

regulatory_handoff

Does the gene's globally-dominant regulator differ between healthy and tumour OR between NAT and tumour. AUTHORED

GENE BOOLEAN 100%

MEAN_ 1 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung gene 1 coverage 82%82% (median 82%)
1x real, signed

mean_dTarget

A per-row mean of a lower-rung quantity; check agg_of for which rung it summarises. AUTHORED

GENE REAL, SIGNED 82%

CONCENTRATION_ 1 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung gene 1 coverage 53%53% (median 53%)
1x fraction 01

concentration_adj

$(\text{concentration} \wedge 1/n_{fam}) / (1 \wedge 1/n_{fam}) \wedge [0,1]$ how concentrated the gene is RELATIVE to the most diffuse its own design allows. AUTHORED

GENE FRACTION 01 53%

FAM_ 1

reference context, provenance and rarely-quoted detail

1	columns	1	rung	gene	1	coverage	81%81% (median 81%)
1x	signed	1	1				

fam_pooled_rho_adj

GENE

SIGNED 11

81%

Seed-family structure the arm sits in membership, affinity and dominance within it.

AUTHORED

gene_family_card.tsv

gene_family

One row per one (seed family, gene) pair. Keyed [gene, family] it cannot express a property of the family itself. That is the seed_family card's job.

61 columns 5,117 rows 12 blocks median coverage 97%

Contents 12 blocks, most important first

1. (bare) 8 CORE

2. n_3 SUPPORTING

3. ctx_16 SUPPORTING

4. beta_7 SUPPORTING

5. comp_16 REFERENCE

6. identity_4 REFERENCE
7. pip_2 REFERENCE

8. w_1 REFERENCE

9. m_1 REFERENCE

10. composition_1 REFERENCE

11. prior_1 REFERENCE

12. retention_1 REFERENCE

(BARE) 8

core the columns findings rest on

8 columns spans family, key this prefix groups columns living on DIFFERENT units coverage 95%100% (median 100%)
3x identifier 3x real 0 1x boolean 1x real, signed

arms FAMILY IDENTIFIER 100%	The arms collapsed into this (gene, family) cell, semicolon-separated (hsa-miR-17-5p; hsa-miR-20a-5p). AUTHORED
beta FAMILY REAL 0 100%	FAMILY-rung coupling coefficient readouts.run(level='family') , the same-seed collapse APPLIED. AUTHORED
family KEY IDENTIFIER 100%	KEY seed family of the regulator, as used by the family-rung fit. AUTHORED
gene KEY IDENTIFIER 100%	KEY HGNC gene symbol. AUTHORED
identified FAMILY BOOLEAN 100%	z > 2 on the UNCALIBRATED SD 27.1% of cells. cal_identified is the honest version. AUTHORED
identity FAMILY REAL, SIGNED 95%	Shapley/LMG identity at the FAMILY rung same estimator as the edge card's identity , different unit. AUTHORED
retention FAMILY REAL 0 100%	beta_deconv / beta_core at the family grain the ADJUSTMENT sense of retention (what fraction of the coupling survives the composition block), never the patient-baseline sense. AUTHORED
z FAMILY REAL 0 100%	beta / beta_sd for the family cell, on the UNCALIBRATED posterior SD. AUTHORED

N_ 3

supporting read these when the core raises a question

3 columns spans family, gene this prefix groups columns living on DIFFERENT units coverage 100%100% (median 100%)
2x count 1x constant

n_arms

FAMILY

COUNT

100%

How many arms of THIS family sit in THIS gene's design. FAMILY rung, VERIFIED (2026-08-19): it varies within gene for 241 of 1,549 genes, and the per-gene SUM of these equals the gene card's n_arms exactly a clean cross-rung consistency check. AUTHORED

n_fam

GENE

COUNT

100%

How many families compete at this gene. beta is mathematically INERT at 1. AUTHORED

n_samples

GENE

CONSTANT

100%

Samples the family cell was fit on CONSTANT 1,040 across the card. AUTHORED

CTX_ 16

supporting read these when the core raises a question

Gene-design context from the decoy/atlas layer (learned/analyses/gene_atlas.py via card_context): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

16 columns rung gene SUBSET of the edge block, identical rungs coverage 81%100% (median 96%) 13 share the block description
4x signed 1 3x real 0 3x real, signed 3x count 1x categorical 1x fraction 01 1x boolean

ctx_apriori_class

GENE

CATEGORICAL

100%

class label DERIVED

ctx_ceiling

GENE

SIGNED 1

99%

The UPPER BOUND on what ANY estimator can achieve for this gene the 5-fold out-of-fold R^2 of an UNRESTRICTED OLS on the gene's real design (card_context.py:15). AUTHORED

ctx_collapse

GENE

REAL 0

96%

n_arm / n_fam for the gene how much curation BUNDLES same-seed mates onto it (15). AUTHORED

ctx_d_collin

GENE

SIGNED 1

81%

change DERIVED
Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)

ctx_d_fam

GENE

REAL, SIGNED

95%

change per seed family DERIVED

ctx_dose_max

GENE

REAL 0

96%

dose maximum DERIVED

ctx_dose_med

GENE

REAL 0

96%

dose median DERIVED

ctx_frac_abund

GENE

FRACTION 01

96%

fraction unweighted abundance-sum reference DERIVED

ctx_gap_core

GENE

SIGNED 1

95%

gap core confounder block DERIVED

ctx_gap_d_dose	gap · change · dose	DERIVED
GENE	REAL, SIGNED	95%
ctx_gap_deconv	gap · composition-adjusted (deconvolution block)	DERIVED
GENE	SIGNED $\hat{=} 1$	95%
ctx_gap_retention	gap · retention	DERIVED
GENE	REAL, SIGNED	92%
ctx_measurable	Is the gene's ctx_ceiling above CEILING_ROBUST (0.02, not 0 · MH-144)?	AUTHORED
GENE	BOOLEAN	99%
ctx_n_abund	count · unweighted abundance-sum reference	DERIVED
GENE	COUNT	96%
ctx_n_fam_abund	count · per seed family · unweighted abundance-sum reference	DERIVED
GENE	COUNT	96%
ctx_n_fam_multi	count · per seed family	DERIVED
GENE	COUNT	96%

BETA_ · 7

supporting · read these when the core raises a question

Posterior summaries of the coupling coefficient from the dense Gibbs fit (`learned/readouts.py`). `_sd` is the posterior SD, `_frac` the share of the gene's total beta.

7 columns · `rung` **family** · coverage **100%·100%** (median 100%) · **7** share the block description
4× fraction $\hat{=} 1$ · 3× real $\hat{=} 0$

beta_deconv	composition-adjusted (deconvolution block)	DERIVED
FAMILY	REAL $\hat{=} 0$	100%
beta_frac	fraction	DERIVED
FAMILY	FRACTION $\hat{=} 1$	100%
beta_frac_abs	fraction	DERIVED
FAMILY	FRACTION $\hat{=} 1$	100%
beta_frac_deconv	fraction · composition-adjusted (deconvolution block)	DERIVED
FAMILY	FRACTION $\hat{=} 1$	100%
beta_frac_sd	fraction · standard deviation	DERIVED
FAMILY	FRACTION $\hat{=} 1$	100%
beta_sd	standard deviation	DERIVED
FAMILY	REAL $\hat{=} 0$	100%
beta_sd_deconv	standard deviation · composition-adjusted (deconvolution block)	DERIVED
FAMILY	REAL $\hat{=} 0$	100%

Cell-composition drivers of this gene's coupling (compartment_*): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.

16 columns rung gene coverage 66%97% (median 81%) 7 share the block description
6× categorical 6× signed 1 3× real, signed 1× real 0

comp_cptac_prot_driver	CPTAC against protein DERIVED
GENE CATEGORICAL 66%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_driver_ret	1 comp_cptac_prot_driver_share , exactly. Median +0.398 raw, +0.469 gated to I_raw % ¥ 0.10 “ quote the gated value. See the mRNA twin. AUTHORED
GENE REAL, SIGNED 66%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_driver_share	Protein-layer twin of comp_tcga_mrna_driver_share . Same exact complementarity with _driver_ret (8.9e-16 on 918 rows). AUTHORED
GENE REAL, SIGNED 66%	
comp_cptac_prot_rho_raw	Unadjusted aggregate PROTEIN coupling “ denominator of comp_cptac_prot_driver_ret . AUTHORED
GENE SIGNED 1 81%	
comp_cptac_prot_top_budget	CPTAC against protein the top-ranked pressure budget DERIVED
GENE CATEGORICAL 81%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_budget_r	CPTAC against protein the top-ranked pressure budget correlation DERIVED
GENE SIGNED 1 81%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_target	CPTAC against protein the top-ranked DERIVED
GENE CATEGORICAL 81%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_target_r	CPTAC against protein the top-ranked correlation DERIVED
GENE SIGNED 1 81%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_tcga_mrna_driver	WHICH compartment drives the gene's mRNA coupling “ the one whose removal moves it most (CAFs 279 purity 174 prolif 148 Endothelial 122 Myeloid 109). AUTHORED
GENE CATEGORICAL 81%	
comp_tcga_mrna_driver_ret	1 comp_tcga_mrna_driver_share , exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. AUTHORED
GENE REAL, SIGNED 81%	
comp_tcga_mrna_driver_share	Share of the gene's mRNA coupling attributable to the single compartment that most changes it “ axiom 8's composition GATE, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. AUTHORED
GENE REAL 0 81%	
comp_tcga_mrna_rho_raw	The gene's UNADJUSTED aggregate mRNA coupling “ the DENOMINATOR of comp_tcga_mrna_driver_ret . AUTHORED
GENE SIGNED 1 97%	
comp_tcga_mrna_top_budget	The compartment the gene's miRNA BUDGET loads on most “ a property of the REGULATORS. AUTHORED
GENE CATEGORICAL 97%	
comp_tcga_mrna_top_budget_r	TCGA the top-ranked pressure budget correlation DERIVED
GENE SIGNED 1 97%	

comp_tcga_mrna_top_target

The compartment the TARGET GENE's expression loads on most.

AUTHORED

GENE

CATEGORICAL

97%

comp_tcga_mrna_top_target_r

TCGA the top-ranked correlation

DERIVED

GENE

SIGNED -1;1

97%

IDENTITY_ 4

reference context, provenance and rarely-quoted detail

4 columns rung family coverage 95%97% (median 95%)
2x fraction 01 1x real, signed 1x boolean

identity_abs

|identity| / Î|identity| per gene the BOUNDED companion to identity, always in [0,1] (verified: 0 non-NaN values outside it on either card).

AUTHORED

FAMILY

FRACTION 01

95%

identity_coherence

1 / Î|identity| per GENE, in (0,1] identity's own sign coherence. 1.0 when nothing cancels, 0 as suppressor cancellation grows.

AUTHORED

FAMILY

FRACTION 01

95%

identity_deconv

Shapley/LMG attribution on R^2 with bagged-NNLS weights WHO regulates the gene, collinearity-fair.

AUTHORED

FAMILY

REAL, SIGNED

97%

identity_reliable

identity_coherence 0.5 AND |identity| 1. Admits 93.3% of edge rows and 92.6% of gene_family rows, and excludes every row whose |identity| exceeds 1 (119/119 and 118/118).

AUTHORED

FAMILY

BOOLEAN

95%

PIP_ 2

reference context, provenance and rarely-quoted detail

2 columns rung family coverage 100%100% (median 100%)
1x constant 1x fraction 01

pip_dense

CONSTANT by construction (Î1 dense readout).

AUTHORED

FAMILY

CONSTANT

100%

pip_discovery

Posterior inclusion probability. pip_dense comes from the pi=1 coupling readout; pip_discovery from the evidence-pi readout.

AUTHORED

FAMILY

FRACTION 01

100%

W_ 1

reference context, provenance and rarely-quoted detail

1 columns rung gene coverage 100%100% (median 100%)
1x real 0

w_max

Maximum curated EVIDENCE weight for this (gene, family) cell. Family-rung here; gene-rung on the gene card.

AUTHORED

GENE

REAL 0

100%

M_ 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung family 1 coverage 100%100% (median 100%)

1x real 0

m_nnlsm Bagged-NNLS weight for this (gene, family) cell. AUTHORED

FAMILY

REAL 0

100%

COMPOSITION_ 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung family 1 coverage 100%100% (median 100%)

1x categorical

composition_classComposition-adjustment verdict for this row (e.g. composition_explained when the effect does not survive the deconvolution block). AUTHORED

FAMILY

CATEGORICAL

100%

PRIOR_ 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung family 1 coverage 100%100% (median 100%)

1x fraction 01

prior_piprior_piThe evidence prior fed to the discovery readout. AUTHORED

FAMILY

FRACTION 01

100%

RETENTION_ 1

reference " context, provenance and rarely-quoted detail

1 columns 1 rung family 1 coverage 100%100% (median 100%)

1x boolean

retention_reliableretention_reliableFraction of an effect surviving covariate adjustment (adjusted / raw). AUTHORED

FAMILY

BOOLEAN

100%

arm_card.tsv

arm

One row per the miRNA arm itself, gene-free. The deepest card. An arm-rung column must be constant within arm ACROSS genes; if it varies by gene it is not an arm property, whatever its name says.

296 columns 2,450 rows 44 blocks median coverage 30%

Contents 44 blocks, most important first

1. adm_4

2. arm_7

3. seed_1

4. healthy_3

5. site_12

6. arb_16

7. fame_28

8. (bare)_1

9. ago_4

10. dose_3

11. abund_5

12. hly_7

13. ctx_arm_2

14. attr_14

15. grank_3

16. real_11

17. ts_10

18. isoc_5

19. chim_5

20. sdsz_4

21. dGlobal_3

22. fst_16
23. seq_4

24. sub_6

25. cov_22

26. cur_he_2

27. iso_5

28. field_5

29. famrole_6

30. bc_11

31. gctx_16

32. surv_8

33. aid_4

34. wshift_6

35. tier_5

36. surrogate_2

37. disc_2

38. foc_7

39. cnv_9

40. cnvc_5

41. model_1

42. comv_2

43. cload_3

44. hx_1
23. seq_4

24. sub_6

25. cov_22

26. cur_he_2

27. iso_5

28. field_5

29. famrole_6

30. bc_11

31. gctx_16

32. surv_8

33. aid_4

34. wshift_6

35. tier_5

36. surrogate_2

37. disc_2

38. foc_7

39. cnv_9

40. cnvc_5

41. model_1

42. comv_2

43. cload_3

44. hx_1

ADM_ 4

core the columns findings rest on

4 columns Â· rung arm Â· AGGREGATE of the edge block Â· verified: n_with_site and n_admissible reproduce a direct rollup on 100% of arms (n_edges only 77.5%, the MH-252 universe gap) Â· coverage 14%29% (median 29%)
3× count Â· 1× fraction 01

adm_frac_with_site

ARM FRACTION 01 14%

adm_n_with_site / adm_n_edges .

AUTHORED

adm_n_admissible

ARM COUNT 29%

Edges of this arm that pass admissibility (adm_has_site + the evidence bar) Â· MH-216's single most load-bearing conditioning variable, rolled up per arm.

AUTHORED

adm_n_edges

ARM COUNT 29%

ADMISSIBILITY Â· could this edge repress in breast AT ALL? has_site (seed match in the 3'UTR) AND expressed (arm present in TCGA tumour OR GTEx healthy breast).

AUTHORED

adm_n_with_site ARM COUNT 29%	Edges of this arm whose target carries a seed site â€” 89.9% of edges (4,436/4,937) , so the site requirement is NOT the binding constraint; the evidence bar is (admissible drops it to 60.4%). 19.7% of arms have none. AUTHORED Defined on: edges/arms with an admissibility tag (4,937 edges; has_site 89.9% Â· expressed 63.9% Â· admissible 60.4%)
---	---

ARM_ Â· 7
core â€” the columns findings rest on

7 columns Â· rung arm Â· coverage 15%â€”20% (median 20%) 4x real, signed Â· 2x real â‰¥ 0 Â· 1x percent

arm_iqr ARM REAL â‰¥ 0 20%	Interquartile range of the arm's tumour RPM â€” its DYNAMIC RANGE. AUTHORED
arm_lfc_HLY_NAT_raw ARM REAL, SIGNED 15%	log-FC GTEx-healthy â†’ TCGA-NAT, raw. AUTHORED
arm_lfc_HLY_TUM_QN ARM REAL, SIGNED 16%	log-FC GTEx-healthy â†’ TCGA-tumour, on quantile-normalised values. AUTHORED
arm_lfc_HLY_TUM_raw ARM REAL, SIGNED 15%	As arm_lfc_HLY_TUM_QN but against the RAW GTEx median. AUTHORED
arm_lfc_NAT_TUM ARM REAL, SIGNED 20%	log-FC of the arm's median abundance, matched NAT â†’ tumour. AUTHORED
arm_med_rpm ARM REAL â‰¥ 0 20%	The arm's MEDIAN RPM across tumour samples â€” its abundance level, gene-free. Defined on the 19.8% of arms with tumour expression data. AUTHORED Defined on: edges present in the progression panel (arm expressed + state measured)
arm_pct_above_floor ARM PERCENT 20%	â€” THE NAME READS BACKWARDS . Computed as <code>100Â·mean(x > FLOOR)</code> â€” the percent of samples ABOVE the detection floor, i.e. AUTHORED

SEED_ Â· 1
core â€” the columns findings rest on

1 columns Â· rung arm Â· coverage 100%â€”100% (median 100%) 1x identifier

seed_family ARM IDENTIFIER 100%	KEY â€” the seed family itself, one row per family. AUTHORED
---	---

HEALTHY_ Â· 3
core â€” the columns findings rest on

3 columns Â· rung arm Â· identical column set to the edge block Â· coverage 16%â€”20% (median 16%) 1x categorical Â· 1x real â‰¥ 0 Â· 1x boolean
--

healthy_leg ARM CATEGORICAL 20%	Provenance of this arm's GTEx healthy leg â€” <code>measured</code> , <code>collapse_blind</code> or <code>true_absent</code> . AUTHORED
---	---

healthy_potential The arm's IMPUTED healthy abundance $\hat{\epsilon}$ a repression-POTENTIAL proxy, collapse-repaired (0.27 $\hat{\epsilon}$ 17.9). **AUTHORED**

healthy_uninformative Is the healthy leg both `collapse_blind` AND high-potential $\hat{\epsilon}$ i.e. the one combination where an *acquired* call is genuinely unsafe? **True on 143 of 3,490 edges (4%).** **AUTHORED**

ARB_ 16

supporting read these when the core raises a question

Per-ARM rollup of its edge set how many edges and genes it regulates and its mean |beta| over them.

16 columns rung arm coverage 28%30% (median 30%) 10 share the block description

12x count 2x real 1x fraction 01 1x signed 11

arb_frac_identified ARM FRACTION 01 AGG EDGE 30%	arb_n_identified / arb_n_edges (verified exact on 100% of rows). AUTHORED
arb_max_abs_beta ARM REAL 0 AGG EDGE 30%	Largest $\hat{I}^2$ among the arm's edges. AUTHORED
arb_max_identity ARM SIGNED 11 AGG EDGE 28%	Largest identity among this arm's edges, over reliable rows only. Now runs 0.255 +1.000 it reached +740.0 until the inert beta_frac_reliable gate was replaced (2026-08-19). AUTHORED
arb_mean_abs_beta ARM REAL 0 AGG EDGE 30%	Mean $\hat{I}^2$ over the arm's edges. See arb_max_abs_beta for the one-edge degeneracy (9.3% of arms, where the two are equal by construction). AUTHORED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b001 ARM COUNT 30%	count at the 0.001 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b001_ident ARM COUNT 30%	count at the 0.001 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b005 ARM COUNT 30%	count at the 0.005 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b005_ident ARM COUNT 30%	count at the 0.005 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b010 ARM COUNT 30%	count at the 0.010 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b010_ident ARM COUNT 30%	count at the 0.010 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b025 ARM COUNT 30%	count at the 0.025 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b025_ident ARM COUNT 30%	count at the 0.025 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_comp_explained ARM COUNT AGG EDGE 30%	count DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_edges ARM COUNT AGG EDGE 30%	Edges this arm carries in the LEARNED MODEL's own readouts. AUTHORED
arb_n_identified ARM COUNT AGG EDGE 30%	count DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)

arb_n_identity_reliable	Edges of this arm whose identity share is usable (identity_reliable).	AUTHORED
ARM	COUNT	AGG EDGE 30%

FAME_ 28
supporting “ read these when the core raises a question

Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.

28 columns Â· rung arm Â· coverage 94%96% (median 96%) Â· 23 share the block description

27x count Â· 1x fraction 0â€”1

fame_assay_binding_functional_mti_weak_studies	by assay class Â· functional assays Â· miRNA“target interaction evidence Â· weak evidence Â· study count	DERIVED	Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
ARM	COUNT	96%	
fame_assay_binding_studies	Studies whose assay class is BINDING (the marginal of the assay – functional-class cross-tab).	AUTHORED	
ARM	COUNT	96%	
fame_assay_functional_mti_studies	by assay class Â· functional assays Â· miRNA“target interaction evidence Â· study count	DERIVED	Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
ARM	COUNT	96%	
fame_assay_functional_mti_weak_studies	by assay class Â· functional assays Â· miRNA“target interaction evidence Â· weak evidence Â· study count	DERIVED	Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
ARM	COUNT	96%	
fame_assay_nonfunctional_mti_studies	by assay class Â· miRNA“target interaction evidence Â· study count	DERIVED	Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
ARM	COUNT	96%	
fame_assay_other_functional_mti_studies	by assay class Â· functional assays Â· miRNA“target interaction evidence Â· study count	DERIVED	Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
ARM	COUNT	96%	
fame_assay_other_functional_mti_weak_studies	by assay class Â· functional assays Â· miRNA“target interaction evidence Â· weak evidence Â· study count	DERIVED	Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
ARM	COUNT	96%	
fame_assay_other_nonfunctional_mti_studies	by assay class Â· miRNA“target interaction evidence Â· study count	DERIVED	Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
ARM	COUNT	96%	
fame_assay_other_studies	by assay class Â· study count	DERIVED	Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
ARM	COUNT	96%	
fame_assay_protein_functional_mti_studies	by assay class Â· functional assays Â· miRNA“target interaction evidence Â· study count	DERIVED	Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
ARM	COUNT	96%	
fame_assay_protein_functional_mti_weak_studies	by assay class Â· functional assays Â· miRNA“target interaction evidence Â· weak evidence Â· study count	DERIVED	Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
ARM	COUNT	96%	
fame_assay_protein_nonfunctional_mti_studies	by assay class Â· miRNA“target interaction evidence Â· study count	DERIVED	Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
ARM	COUNT	96%	

fame_assay_protein_studies ARM COUNT 96%	by assay class $\hat{A}$ · study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_assay_reporter__functional_mti_studies ARM COUNT 96%	by assay class $\hat{A}$ · functional assays $\hat{A}$ · miRNA $\hat{a}$ €“target interaction evidence $\hat{A}$ · study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_assay_reporter__nonfunctional_mti_studies ARM COUNT 96%	by assay class $\hat{A}$ · miRNA $\hat{a}$ €“target interaction evidence $\hat{A}$ · study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_assay_reporter_studies ARM COUNT 96%	by assay class $\hat{A}$ · study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_assay_rna__functional_mti_studies ARM COUNT 96%	by assay class $\hat{A}$ · against mRNA $\hat{A}$ · functional assays $\hat{A}$ · miRNA $\hat{a}$ €“target interaction evidence $\hat{A}$ · study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_assay_rna__functional_mti_weak_studies ARM COUNT 96%	by assay class $\hat{A}$ · against mRNA $\hat{A}$ · functional assays $\hat{A}$ · miRNA $\hat{a}$ €“target interaction evidence $\hat{A}$ · weak evidence $\hat{A}$ · study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_assay_rna__nonfunctional_mti_studies ARM COUNT 96%	by assay class $\hat{A}$ · against mRNA $\hat{A}$ · miRNA $\hat{a}$ €“target interaction evidence $\hat{A}$ · study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_assay_rna_studies ARM COUNT 96%	by assay class $\hat{A}$ · against mRNA $\hat{A}$ · study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_assay_total_studies ARM COUNT 96%	Distinct STUDIES (<code>study_id.nunique()</code>) behind this arm's curated edges $\hat{a} \in$ ” median 57, max 930. See <code>fame_assay_unique_targets</code> : different definitions, one axis (spearman 1.0000). AUTHORED
fame_assay_unique_targets ARM COUNT 96%	Distinct TARGET GENES this arm has in the curated literature (<code>gene.nunique()</code>). AUTHORED
fame_led_frac_ltfunc ARM FRACTION 0$\hat{A}$€“1 AGG EDGE 96%	fraction DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_led_n_classes ARM COUNT AGG EDGE 96%	How many distinct evidence CLASSES this arm's curated edges span (1 $\hat{A}$ €“7). AUTHORED
fame_led_n_ltfunc_pmid ARM COUNT AGG EDGE 96%	count $\hat{A}$ · distinct publications DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_led_n_weak ARM COUNT AGG EDGE 94%	count $\hat{A}$ · weak evidence DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_n_genes_curated ARM COUNT AGG EDGE 96%	count $\hat{A}$ · genes DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{a} \in \hat{a}$ ” FAME axes, not targetome (MH-208)
fame_npmid ARM COUNT AGG EDGE 96%	Distinct PMIDs citing this arm. AUTHORED

(BARE) 1

supporting read these when the core raises a question

1 columns

run key

coverage 100%100% (median 100%)

1x identifier

arm

KEYIDENTIFIER100%

KEY read mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()`, which gates against the canonical universe and LOGS drops rather than silently dropping. AUTHORED

AGO_ 4

supporting read these when the core raises a question

4 columns

run arm

arm.ago_dom and edge.ago_loading are RANK-IDENTICAL (spearman +1.000, equal on 73.2%) one quantity, two names. Never scan both.

coverage 28%60% (median 28%)

1x fraction 01 1x constant 1x categorical 1x real 0

ago_dom

ARMFRACTION 0128%

Fraction of the arm's Manakov AGO-CLIP reads assigned to it rather than its sibling arm the loading dominance. Median 0.658 on the 27.7% of arms with CLIP coverage. AUTHORED

ago_dom_src

ARMCONSTANT28%

Provenance of `ago_dom` which CLIP dataset the AGO-loading dominance came from. AUTHORED

ago_guide_class

ARMCATEGORICAL28%

AGO/RISC loading gate for the arm. AUTHORED

ago_reads

ARMREAL 060%

Raw AGO-CLIP read count (median 6, max 210,722). AUTHORED

DOSE_ 3

supporting read these when the core raises a question

Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic.

3 columns

run arm

SUBSET of the edge block

coverage 12%12% (median 12%)

2 share the block description

2x real 0 1x constant

dose_comp_retention

ARMREAL 012%

retention DERIVED

Defined on: edges with a matched NAT leg (n~104 paired participants)

dose_confounded

ARMCONSTANT12%

Is this arm's NAT's tumour dose a composition or proliferation artifact (either gated retention < 0.4)? AUTHORED

dose_prolif_retention

ARMREAL 012%

retention DERIVED

Defined on: edges with a matched NAT leg (n~104 paired participants)

ABUND_ 5

supporting read these when the core raises a question

The arm's own abundance profile across the cohort median, SD, IQR, detection fraction.

5 columns : rung arm : coverage 88%91% (median 91%) : 4 share the block description
3x real : 0 : 1x fraction 01 : 1x categorical

abund_frac_detected	fraction	DERIVED	
ARM	FRACTION 01	91%	Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
abund_iqr	interquartile range	DERIVED	
ARM	REAL 0	91%	Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
abund_med	median	DERIVED	
ARM	REAL 0	91%	Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
abund_sd	standard deviation	DERIVED	
ARM	REAL 0	88%	Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
abund_sd_tertile	Dispersion tertile read 718 / 716 / 718 by construction, so it carries no information beyond abund_sd's rank.	AUTHORED	
ARM	CATEGORICAL	88%	

HLY_ 7

supporting read these when the core raises a question

HEALTHY baseline levels for the arm read GTEx breast median/percentile and TCGA-NAT median.

7 columns : rung arm : coverage 17%91% (median 91%) : 3 share the block description
2x real : 0 : 1x categorical : 1x boolean : 1x percent : 1x count : 1x fraction 01

hly_baseline_src	source	DERIVED	
ARM	CATEGORICAL	91%	Defined on: arms with a GTEx healthy anchor read MEASURED leg only; floor0 is NEVER emitted as a value
hly_from_seedmate	Was this arm's healthy baseline borrowed from a SEED-MATE rather than measured directly (49 arms)?	AUTHORED	
ARM	BOOLEAN	91%	
hly_gtex_median	GTEx read median	DERIVED	
ARM	REAL 0	17%	Defined on: arms with a GTEx healthy anchor read MEASURED leg only; floor0 is NEVER emitted as a value
hly_gtex_pct	GTEx read percent	DERIVED	
ARM	PERCENT	17%	Defined on: arms with a GTEx healthy anchor read MEASURED leg only; floor0 is NEVER emitted as a value
hly_nat_detect_n	In how many of the ~104 matched NAT samples this arm is detected above the floor.	AUTHORED	
ARM	COUNT	91%	
hly_nat_median	Median NAT abundance (fill 74.0%).	AUTHORED	
ARM	REAL 0	74%	
hly_tumor_prev	Prevalence of the arm above the detection floor in tumour (median 0.01).	AUTHORED	
ARM	FRACTION 01	91%	

CTX_ARM_ 2

reference “ context, provenance and rarely-quoted detail

2 columns • rung arm • the two ARM-RUNG members of ctx_, which the other cards do not carry • coverage 29%“29% (median 29%)
1× boolean • 1× real %¥ 0

ctx_arm_abundant	Is ctx_arm_dose >= log2(11) , i.e. is this arm abundant enough to plausibly repress? Median 0.00 “ most arms are NOT.
ARM	BOOLEAN
29%	AUTHORED

ctx_arm_dose	This arm's median log2(RPM+1) across tumour samples (0.13 “ 17.92, median 1.08).
ARM	REAL %¥ 0
29%	AUTHORED

ATTR_ 14

reference “ context, provenance and rarely-quoted detail

Attribution rollup over the arm's edges “ how much of the gene's coupling this arm is credited with, and over how many reliable edges.
14 columns • rung arm • coverage 6%“29% (median 15%) • 9 share the block description
11× fraction 0“1 • 2× count • 1× signed ^’1“1

attr_max_budget_share	maximum • pressure budget • share
ARM	FRACTION 0“1
15%	DERIVED
Defined on: arms with a Shapley/attribution decomposition over their edges (728)	

attr_max_credit_share	maximum • share
ARM	FRACTION 0“1
6%	DERIVED
Defined on: arms with a Shapley/attribution decomposition over their edges (728)	

attr_max_joint_share	Largest JOINT attribution share among this arm's edges.
ARM	FRACTION 0“1
15%	AUTHORED

attr_max_nested_share	Largest NESTED attribution share. See attr_max_joint_share : different values (agree on 35.4%), one axis (spearman +0.962).
ARM	FRACTION 0“1
15%	AUTHORED
Defined on: arms with a Shapley/attribution decomposition over their edges (728)	

attr_max_within_share	Largest WITHIN-cell attribution share for this arm.
ARM	FRACTION 0“1
15%	AUTHORED

attr_med_budget_share	median • pressure budget • share
ARM	FRACTION 0“1
15%	DERIVED
Defined on: arms with a Shapley/attribution decomposition over their edges (728)	

attr_med_credit_share	median • share
ARM	FRACTION 0“1
6%	DERIVED
Defined on: arms with a Shapley/attribution decomposition over their edges (728)	

attr_med_dose_share	Median dose share of this arm within its family cells.
ARM	FRACTION 0“1
29%	AUTHORED

attr_med_joint_share	median • share
ARM	FRACTION 0“1
15%	DERIVED
Defined on: arms with a Shapley/attribution decomposition over their edges (728)	

attr_med_nested_share	median • share
ARM	FRACTION 0“1
15%	DERIVED
Defined on: arms with a Shapley/attribution decomposition over their edges (728)	

attr_med_phi_joint	median
ARM	SIGNED ^’1“1
15%	DERIVED
Defined on: arms with a Shapley/attribution decomposition over their edges (728)	

attr_med_within_share ARM FRACTION 0Å€"1 15%	median Å· share DERIVED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_n_edges ARM COUNT 22%	count DERIVED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_n_reliable ARM COUNT 15%	Reliable edges behind this arm's attr_* shares (median 2). AUTHORED

GRANK_ Å· 3

reference Å€" context, provenance and rarely-quoted detail

The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from.

3 columns Å· rung **arm** Å· coverage **15Å€"20%** (median 20%) Å· **3** share the block description
3× count

grank_HLY ARM COUNT 15%	in the GTEx-healthy state DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state Å€" many arms are multi-mapping-collapsed, MH-166)
grank_NAT ARM COUNT 20%	in matched normal (NAT) DERIVED Defined on: edges with a matched NAT leg (n~104 paired participants)
grank_TUM ARM COUNT 20%	in tumour DERIVED Defined on: edges present in the progression panel (arm expressed + state measured)

REAL_ Å· 11

reference Å€" context, provenance and rarely-quoted detail

Realization rollup for the ARM Å€" how many of its edges are scored, and its median and best realized coupling.

11 columns Å· rung **arm** Å· coverage **9Å€"24%** (median 24%) Å· **8** share the block description
7× count Å· 2× signed Å·'1Å€|1 Å· 2× fraction 0Å€"1

real_best_coupling ARM SIGNED Å·'1Å€ 1 24%	Most negative realized coupling among this arm's scored genes (median Å·'0.08, min Å·'0.56). AUTHORED
real_frac_c10 ARM FRACTION 0Å€"1 9%	Fraction of this arm's scored genes coupling below Å·'0.10. AUTHORED
real_frac_sig ARM FRACTION 0Å€"1 9%	fraction DERIVED Defined on: arms with Å·%¥1 calibrated-coupling-scored edge (593) Å€" the realization LADDER; rates are gated at nÅ·%¥5
real_med_coupling ARM SIGNED Å·'1Å€ 1 24%	median DERIVED Defined on: arms with Å·%¥1 calibrated-coupling-scored edge (593) Å€" the realization LADDER; rates are gated at nÅ·%¥5
real_n_c05 ARM COUNT 24%	count Å· at the Å·'0.05 threshold DERIVED Defined on: arms with Å·%¥1 calibrated-coupling-scored edge (593) Å€" the realization LADDER; rates are gated at nÅ·%¥5
real_n_c10 ARM COUNT 24%	count Å· at the Å·'0.10 threshold DERIVED Defined on: arms with Å·%¥1 calibrated-coupling-scored edge (593) Å€" the realization LADDER; rates are gated at nÅ·%¥5

real_n_c15 ARMCOUNT24%	count at the 0.15 threshold DERIVED Defined on: arms with 1 calibrated-coupling-scored edge (593) the realization LADDER; rates are gated at 5
real_n_c20 ARMCOUNT24%	count at the 0.20 threshold DERIVED Defined on: arms with 1 calibrated-coupling-scored edge (593) the realization LADDER; rates are gated at 5
real_n_c30 ARMCOUNT24%	count at the 0.30 threshold DERIVED Defined on: arms with 1 calibrated-coupling-scored edge (593) the realization LADDER; rates are gated at 5
real_n_scored ARMCOUNT24%	Genes scored for this arm's realized coupling (median 3). Its n_c05 n_c30 ladder is properly nested 0 violations (verified). AUTHORED Defined on: arms with 1 calibrated-coupling-scored edge (593) the realization LADDER; rates are gated at 5
real_n_sig ARMCOUNT24%	count DERIVED Defined on: arms with 1 calibrated-coupling-scored edge (593) the realization LADDER; rates are gated at 5

TS_ 10

reference the context, provenance and rarely-quoted detail

TargetScan context++ scores for the arm's sites number of sites, mean and best context score. More negative context++ = stronger predicted repression.

10 columns arm coverage 13% (median 13%) 8 share the block description
5x count 2x real, signed 1x signed 1 1x fraction 1 1x real 0

ts_ctx_best ARMREAL, SIGNED13%	Best (most negative) TargetScan context++ score for this arm range [2.02, 0.34], all negative by construction (context++ scores repression as a negative number). AUTHORED
ts_ctx_mean ARMSIGNED 1113%	mean DERIVED Defined on: arms in TargetScan's human default predictions (321)
ts_frac_8mer ARMFRACTION 113%	fraction 8mer sites DERIVED Defined on: arms in TargetScan's human default predictions (321)
ts_kd_med ARMREAL, SIGNED13%	median DERIVED Defined on: arms in TargetScan's human default predictions (321)
ts_n_7mer_A1 ARMCOUNT13%	count 7mer sites DERIVED Defined on: arms in TargetScan's human default predictions (321)
ts_n_7mer_m8 ARMCOUNT13%	count 7mer sites DERIVED Defined on: arms in TargetScan's human default predictions (321)
ts_n_8mer ARMCOUNT13%	count 8mer sites DERIVED Defined on: arms in TargetScan's human default predictions (321)
ts_n_genes ARMCOUNTAGG EDGE13%	Genes with a TargetScan context++ site median 497, fill 12.9%, the SPARSEST universe on the card. Correlates +0.726 with the MANE seed scan and only +0.231 with scanMiR-any. AUTHORED
ts_n_sites ARMCOUNTAGG EDGE13%	count DERIVED Defined on: arms in TargetScan's human default predictions (321)

ts_sites_per_gene

ARMREAL 013%

per geneDERIVED

Defined on: arms in TargetScan's human default predictions (â€ only 321)

ISOC_ 5

reference context, provenance and rarely-quoted detail

isomiR COMPOSITION what fraction of the arm's reads are canonical, orphan, or donated from a neighbouring arm.
5 columns 1 2 3 4 5
3x fraction 1 1x real 0 1x count

isoc_canonical_frac

ARMFRACTION 191%

canonical sites fractionDERIVED

Defined on: arms in the isomiR seed-COMPOSITION table where the reads go; isoc_orphan_frac flags MH-153 mass loss

isoc_donated_frac

ARMFRACTION 191%

fractionDERIVED

Defined on: arms in the isomiR seed-COMPOSITION table where the reads go; isoc_orphan_frac flags MH-153 mass loss

isoc_med_reads

ARMREAL 091%

Median isomiR reads for this arm median 1.00, max 528,927. AUTHORED

isoc_n_dest_families

ARMCOUNT91%

countDERIVED

Defined on: arms in the isomiR seed-COMPOSITION table where the reads go; isoc_orphan_frac flags MH-153 mass loss

isoc_orphan_frac

ARMFRACTION 191%

fractionDERIVED

Defined on: arms in the isomiR seed-COMPOSITION table where the reads go; isoc_orphan_frac flags MH-153 mass loss

CHIM_ 5

reference context, provenance and rarely-quoted detail

Chimeric-evidence ROLLUP for the arm (aggregated over its edges).
5 columns 1 2 3 4 5
4x count 1x real 0

chim_manakov_n_genes

ARMCOUNTAGG EDGE60%

count genesDERIVED

Defined on: arms with chimeric ligation evidence (1,538) ~90% Manakov/HEK293T, per-source, NEVER pooled

chim_manakov_weight

ARMREAL 0AGG EDGE60%

Manakov chimeric support summed over this arm's edges (median 6, max 210,722). AUTHORED

chim_n_sources

ARMCOUNTAGG EDGE63%

countDERIVED

Defined on: arms with chimeric ligation evidence (1,538) ~90% Manakov/HEK293T, per-source, NEVER pooled

chim_tarbase_n_genes

ARMCOUNTAGG EDGE21%

count genesDERIVED

Defined on: arms with chimeric ligation evidence (1,538) ~90% Manakov/HEK293T, per-source, NEVER pooled

chim_tarbase_weight

ARM

COUNT

AGG EDGE

21%

TarBase chimeric support for the arm (median 8, max 10,255), defined on 20.9% of arms – a third of the Manakov coverage.

AUTHORED

Defined on: arms with chimeric ligation evidence (1,538) – ~90% Manakov/HEK293T, per-source, NEVER pooled

SDSZ_ 4

reference – context, provenance and rarely-quoted detail

MANE 3'UTR seed scan.

4 columns – rung arm – coverage 97%–97% (median 97%) – 2 share the block description

2× count – 1× real – 0 – 1× fraction 0–1

sdsz_N_7mer_plus

ARM

COUNT

AGG EDGE

97%

count – 7mer sites

DERIVED

Defined on: arms in the MANE 3'UTR seed scan – a FOURTH targetome universe, never blend (2,370)

sdsz_N_8mer

ARM

COUNT

AGG EDGE

97%

count – 8mer sites

DERIVED

Defined on: arms in the MANE 3'UTR seed scan – a FOURTH targetome universe, never blend (2,370)

sdsz_N_eff

ARM

REAL – 0

AGG EDGE

97%

Effective seed-site count over the scanned universe (median 2,094, fill 96.7% – the best-covered targetome axis).

AUTHORED

sdsz_rarity

ARM

FRACTION 0–1

97%

The arm's seed RARITY, min-max normalised to 0–1 (median 0.11).

AUTHORED

DGLOBAL_ 3

reference – context, provenance and rarely-quoted detail

Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step.

3 columns – rung arm – coverage 15%–20% (median 15%) – 3 share the block description

3× real, signed

dGlobal_HLY_NAT

ARM

REAL, SIGNED

15%

in the GTEx-healthy state – in matched normal (NAT)

DERIVED

Defined on: arms with a GTEx-HEALTHY leg (the thinnest state – many arms are multi-mapping-collapsed, MH-166)

dGlobal_HLY_TUM

ARM

REAL, SIGNED

15%

in the GTEx-healthy state – in tumour

DERIVED

Defined on: arms with a GTEx-HEALTHY leg (the thinnest state – many arms are multi-mapping-collapsed, MH-166)

dGlobal_NAT_TUM

ARM

REAL, SIGNED

20%

in matched normal (NAT) – in tumour

DERIVED

Defined on: edges with a matched NAT leg (n~104 paired participants)

FST_ 16

reference “ context, provenance and rarely-quoted detail

â THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES â” gene-free (`card_ladders.family_state_shift` , 2026-08-19).

16 columns Â· `rung arm` Â· coverage **3%â€”100%** (median 20%) Â· **7** share the block description
6× count Â· 5× boolean Â· 3× fraction 0â€”1 Â· 2× signed âˆ’1â€”1

fst_d_share_HLY_TUM

ARM **SIGNED** **âˆ’1â€”1** **3%**

As `fst_d_share_NAT_TUM` but GTEx-healthy â†’ tumour, same common-support correction. n=68 (HLY is the sparsest leg at 17.0%). Post-fix **mean âˆ’0.0000, 47.1% positive, p=0.93.** **AUTHORED**

fst_d_share_NAT_TUM

ARM **SIGNED** **âˆ’1â€”1** **6%**

Change in the arm's share of its seed family's linear dose, matched NAT â†’ tumour, **re-normalised over the members measured in BOTH states.** **AUTHORED**

fst_dominance_switch

ARM **BOOLEAN** **7%**

Did the family's DOMINANT arm change between healthy and tumour (103 true / 74 false)? **AUTHORED**

fst_hly_family_measured

ARM **BOOLEAN** **100%**

Do ALL members of this arm's family have a measured GTEx level. **AUTHORED**

fst_hly_measured

ARM **BOOLEAN** **100%**

in the GTEx-healthy state **DERIVED**

fst_is_dominant_HLY

ARM **BOOLEAN** **17%**

the dominant one Â· in the GTEx-healthy state **DERIVED**

fst_is_dominant_TUM

ARM **BOOLEAN** **20%**

the dominant one Â· in tumour **DERIVED**

fst_n_common_HLY_TUM

ARM **COUNT** **9%**

Members measured in BOTH GTEx-healthy and tumour â€” the denominator behind `fst_d_share_HLY_TUM`. See `fst_n_common_NAT_TUM`. **AUTHORED**

fst_n_common_NAT_TUM

ARM **COUNT** **20%**

How many of the family's arms are measured in BOTH NAT and tumour â€” the DENOMINATOR behind `fst_d_share_NAT_TUM`, shipped because the column was wrong for want of it. **AUTHORED**

fst_n_members

ARM **COUNT** **100%**

count Â· family members **DERIVED**

fst_rank_HLY

ARM **COUNT** **17%**

rank Â· in the GTEx-healthy state **DERIVED**

fst_rank_NAT

ARM **COUNT** **74%**

rank Â· in matched normal (NAT) **DERIVED**

fst_rank_TUM

ARM **COUNT** **20%**

rank Â· in tumour **DERIVED**

fst_share_HLY

ARM **FRACTION** **0â€”1** **17%**

Share of family dose in GTEx-healthy, from `hly_gtex_median` (fill **17.0%** â€” the sparsest leg). Same singleton degeneracy; and cross-platform, so contrast only. **AUTHORED**

fst_share_NAT

ARM **FRACTION** **0â€”1** **74%**

Share of family dose in matched NAT, from `hly_nat_median` (fill **74.0%** â€” the best-covered leg, 3.7Â— TUM). Same singleton degeneracy: median 1.000, mask on `fst_n_members > 1.` **AUTHORED**

fst_share_TUM

ARM

FRACTION 0.1

20%

The arm's fraction of its seed family's total linear dose in tumour, from arm_med_rpm (fill 19.8%). AUTHORED

SEQ_ 4

reference context, provenance and rarely-quoted detail

4 columns

rung arm

coverage 30%30% (median 30%)

2x count

2x real 0

seq_n_genes_any

ARM

COUNT

AGG EDGE

30%

Genes with ANY scanMiR K_D site AUTHORED

seq_n_genes_strong

ARM

COUNT

AGG EDGE

30%

Genes with a STRONG scanMiR K_D site median 3,632, spearman +0.822 with sdsz_N_eff and +0.797 with site_n_genes_canonical , but only +0.566 with ts_n_genes . AUTHORED

seq_promisc

ARM

REAL 0

30%

log-scale promiscuity over STRONG scanMiR sites (4.45 8.94). AUTHORED

seq_promisc_any

ARM

REAL 0

30%

Promiscuity over ANY scanMiR site (6.46 9.45) the permissive cut. AUTHORED

SUB_ 6

reference context, provenance and rarely-quoted detail

PAM50 subtype block aggregated to the ARM. See esub_ for the per-edge form.

6 columns

rung arm

coverage 30%30% (median 30%)

4 share the block description

2x count

2x signed 1

1x categorical

1x fraction 0.1

sub_best_frac

ARM

FRACTION 0.1

30%

the best fraction DERIVED

Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) DISTRIBUTIONAL (MH-165), not a validated label

sub_best_mode

ARM

CATEGORICAL

30%

the best DERIVED

Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) DISTRIBUTIONAL (MH-165), not a validated label

sub_med_rho_best

ARM

SIGNED 1

30%

Median coupling in the arm's BEST subtype. AUTHORED

sub_med_rho_pool

ARM

SIGNED 1

30%

Median pooled-subtype coupling (median 0.0064) the unselected reference for sub_med_rho_best . AUTHORED

Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) DISTRIBUTIONAL (MH-165), not a validated label

sub_n_edges

ARM

COUNT

30%

count DERIVED

Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) DISTRIBUTIONAL (MH-165), not a validated label

sub_n_q05

ARM

COUNT

30%

count DERIVED

Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) DISTRIBUTIONAL (MH-165), not a validated label

COV_ 22

reference “ context, provenance and rarely-quoted detail

Coverage flags “ was this row ever SCANNED by the named source.

22 columns rung arm coverage 100%100% (median 100%) 2 share the block description

22× boolean

cov_ago_dom	Was this arm ever SCANNED for AGO-loading dominance from Manakov CLIP? True on 678 of 2,450 arms. AUTHORED
cov_attr	Was this arm ever SCANNED for the Shapley/NNLS attribution block? True on 546 of 2,450 arms. AUTHORED
cov_beta	$\hat{I}^2$ DERIVED
cov_cnv	Was this arm ever SCANNED for the arm-locus copy-number block? True on 856 of 2,450 arms. AUTHORED
cov_comp	Was this arm ever SCANNED for the compartment-loading (cload_) block? True on 205 of 2,450 arms. AUTHORED
cov_cptac	CPTAC DERIVED Defined on: arms measurable in CPTAC (2,095)
cov_curated	Was this arm ever SCANNED for the curated-literature blocks (fame_ , cur_)? True on 875 of 2,450 arms. AUTHORED
cov_expr	Was this arm ever SCANNED for the expression / abundance blocks? True on 2,231 of 2,450 arms. AUTHORED
cov_famrole	Was this arm ever SCANNED for the family-role block? True on 2,231 of 2,450 arms. AUTHORED
cov_field	Was this arm ever SCANNED for the focal-field correlation block? True on 571 of 2,450 arms. AUTHORED
cov_focal	Was this arm ever SCANNED for the focal-locus CNV block? True on 733 of 2,450 arms. AUTHORED
cov_fst	Was the family-share block computed for this arm at all “ i.e. did it have the abundance input. AUTHORED
cov_gctx	Was this arm ever SCANNED for the genomic-context block? True on 714 of 2,450 arms. AUTHORED
cov_isocomp	Was this arm ever SCANNED for the isomiR-composition block? True on 2,222 of 2,450 arms. AUTHORED
cov_isomir	Was this arm ever SCANNED for the isomiR seed-shift block? True on 687 of 2,450 arms. AUTHORED
cov_meth	Was this arm ever SCANNED for the locus-methylation (bc_meth_) block? True on 343 of 2,450 arms. AUTHORED
cov_real	Was this arm ever SCANNED for the realized-coupling (real_) block? True on 590 of 2,450 arms. AUTHORED

cov_seed_scan ARM BOOLEAN 100%	Was this arm ever SCANNED for the MANE 3'UTR seed scan (<code>site_</code>)? AUTHORED
cov_seq_scan ARM BOOLEAN 100%	Was this arm ever SCANNED for the scanMiR K_D scan (<code>seq_</code>)? AUTHORED
cov_site_types ARM BOOLEAN 100%	Was this arm ever SCANNED for the site-type breakdown? AUTHORED
cov_subtype ARM BOOLEAN 100%	Was this arm ever SCANNED for the PAM50 subtype block (<code>sub_</code>)? AUTHORED
cov_targetscan ARM BOOLEAN 100%	Was this arm ever SCANNED for the TargetScan context++ block (<code>ts_</code>)? AUTHORED

CUR_HE_ 2
reference “ context, provenance and rarely-quoted detail

2 columns rung arm coverage 36%”36% (median 36%) 2x count	
cur_he_degree ARM COUNT AGG EDGE 36%	How many HE genes this arm regulates in the curated design “ the arm's curation degree (median 3, max 125). AUTHORED Defined on: arms with miRTarBase curation / ledger PMIDs “ ” FAME axes, not targetome (MH-208)
cur_he_degree_expr ARM COUNT AGG EDGE 36%	As <code>cur_he_degree</code> but counting only targets above the expression floor. AUTHORED

ISO_ 5
reference “ context, provenance and rarely-quoted detail

isomiR profile for the arm “ 5' shift fraction and seed-shift occurrence. Drives the isomiR-corrected X_fam, which is BUILT but default-OFF (coupling wash, bidirectional). 5 columns rung arm coverage 28%”28% (median 28%) 4 share the block description 2x fraction “1 1x count 1x real “% 0 1x real, signed	
iso_n_samples ARM COUNT 28%	count DERIVED Defined on: arms with a 5'-isomiR seed-shift measurement (687) “ š gate on iso_n_samples, the median is n-DEPENDENT
iso_rpm_summed ARM REAL “% 0 28%	š A SUM ACROSS SAMPLES, not a per-sample RPM “ the name misleads. Max 270,988,199 , which is impossible for a per-million unit; divided by <code>iso_n_samples</code> the scale is sane (median 57 , max 251,381). AUTHORED
iso_seed_shift_frac ARM FRACTION “”1 28%	shift fraction DERIVED Defined on: arms with a 5'-isomiR seed-shift measurement (687) “ š gate on iso_n_samples, the median is n-DEPENDENT
iso_seed_shift_sd ARM FRACTION “”1 28%	shift standard deviation DERIVED Defined on: arms with a 5'-isomiR seed-shift measurement (687) “ š gate on iso_n_samples, the median is n-DEPENDENT
iso_top_shift_nt ARM REAL, SIGNED 28%	the top-ranked shift DERIVED Defined on: arms with a 5'-isomiR seed-shift measurement (687) “ š gate on iso_n_samples, the median is n-DEPENDENT

FIELD_ 5

reference " context, provenance and rarely-quoted detail

Within-patient NAT field-effect fit " is the regulator already established in the patient's adjacent normal before the tumour. Defined on the paired subset.

5 columns rung arm coverage 23%"23% (median 23%) 3 share the block description

4x signed 11 1x count

field_excess_over_perm ARM SIGNED 11 23%	"a" NOT A RETENTION RATIO " a FIFTH sense of the word in this codebase. Measured: it equals field_r_own field_r_perm (exact within rounding on 100% of 571 arms, max residual 0.001), i.e. AUTHORED
field_n ARM COUNT 23%	count DERIVED Defined on: arms with a within-patient NAT field-effect fit (571)
field_r_adj ARM SIGNED 11 23%	correlation composition-adjusted DERIVED Defined on: arms with a within-patient NAT field-effect fit (571)
field_r_own ARM SIGNED 11 23%	correlation DERIVED Defined on: arms with a within-patient NAT field-effect fit (571)
field_r_perm ARM SIGNED 11 23%	The permutation null for field_r_own " median 0.000, range [0.11, +0.09]. AUTHORED

FAMROLE_ 6

reference " context, provenance and rarely-quoted detail

The arm's ROLE inside its seed family " member count, its share of family abundance, and its rank among members.

6 columns rung arm coverage 88%"100% (median 91%) 5 share the block description

3x count 1x fraction 01 1x categorical 1x boolean

famrole_abund_rank ARM COUNT 91%	unweighted abundance-sum reference rank DERIVED Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1
famrole_abund_share ARM FRACTION 01 91%	unweighted abundance-sum reference share DERIVED Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1
famrole_class ARM CATEGORICAL 100%	The arm's role in its seed family " sole on 1,639 of 2,450 (66.9%), dominant 282, co and the rest below. AUTHORED
famrole_is_dominant ARM BOOLEAN 91%	the dominant one DERIVED Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1
famrole_n_members ARM COUNT 100%	count family members DERIVED Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1
famrole_var_rank ARM COUNT 88%	rank DERIVED Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1

BC_ 11

reference " context, provenance and rarely-quoted detail

Breast-cancer locus annotation for the arm " CpG probes, CGI overlap and related epigenetic context at the precursor locus. 11 columns rung arm coverage 14%36% (median 14%) 9 share the block description 4x fraction 01 4x count 1x boolean 1x signed 11 1x categorical		
bc_fam_context_homogeneous	per seed family DERIVED	Defined on: BROADCAST from the seed FAMILY (genomic_context.family_context)
bc_fam_dose_coding_host_fraction	per seed family dose fraction DERIVED	Defined on: BROADCAST from the seed FAMILY (genomic_context.family_context)
bc_fam_dose_cotranscribed_fraction	per seed family dose fraction DERIVED	Defined on: BROADCAST from the seed FAMILY (genomic_context.family_context)
bc_fam_n_distinct_coding_hosts	per seed family count DERIVED	Defined on: BROADCAST from the seed FAMILY (genomic_context.family_context)
bc_fam_n_members	per seed family count family members DERIVED	Defined on: BROADCAST from the seed FAMILY (genomic_context.family_context)
bc_meth_dbeta	Tumour-minus-normal methylation beta at this arm's locus (median 0.03, range 0.320.23). AUTHORED	
bc_meth_direction	Tumour-vs-normal methylation direction at this arm's locus. AUTHORED	
bc_meth_n_cgi	count DERIVED	Defined on: BROADCAST from the pri-miRNA HAIRPIN " 5p and 3p share one promoter and one 12 (354 arms)
bc_meth_n_probes	count DERIVED	Defined on: BROADCAST from the pri-miRNA HAIRPIN " 5p and 3p share one promoter and one 12 (354 arms)
bc_meth_normal_beta	12 DERIVED	Defined on: BROADCAST from the pri-miRNA HAIRPIN " 5p and 3p share one promoter and one 12 (354 arms)
bc_meth_tumour_beta	12 DERIVED	Defined on: BROADCAST from the pri-miRNA HAIRPIN " 5p and 3p share one promoter and one 12 (354 arms)

GCTX_ 16

reference " context, provenance and rarely-quoted detail

Genomic context of the arm's precursor locus (genomic_context.classify_he_arms): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. 16 columns rung arm coverage 24%29% (median 29%) 6 share the block description 6x categorical 4x boolean 3x identifier 1x constant 1x count 1x real 0		
--	--	--

gctx_clustered ARM BOOLEAN 29%	Does this arm's precursor sit in a genomic CLUSTER with other miRNAs (310 true / 404 false)? AUTHORED
gctx_context_mixed ARM BOOLEAN 29%	Do this arm's multiple loci DISAGREE about their genomic context â€” 20 arms of 714? AUTHORED
gctx_coord_source ARM CONSTANT 29%	Provenance of the genomic-context coordinates. AUTHORED
gctx_host ARM IDENTIFIER 27%	The HOST GENE the precursor sits inside (356 distinct; ENSG00000269842 , MEG9 , MEG3 most common). AUTHORED
gctx_host_region ARM IDENTIFIER 27%	WHERE in the host the precursor sits (74 distinct; intron 2 and intron 1 most common, 61 arms each). An intronic miRNA is spliced from its host's pre-mRNA; an exonic one competes with it. AUTHORED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_host_type ARM CATEGORICAL 27%	The host's BIOTYPE â€” protein_coding 364 Â· lncRNA 303, plus 5 rarer levels. AUTHORED
gctx_in_exon ARM BOOLEAN 29%	Is the precursor inside a host EXON (218 true / 496 false) rather than an intron? AUTHORED
gctx_mir_class ARM CATEGORICAL 29%	class label DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_mirgenedb ARM BOOLEAN 29%	Is this arm in MirGeneDB â€” the curated high-confidence miRNA set (526 true / 188 false)? AUTHORED
gctx_n_loci ARM COUNT 29%	count DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_promoter_class ARM CATEGORICAL 29%	class label DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_promoter_coord_class ARM CATEGORICAL 29%	class label DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_promoter_gene ARM IDENTIFIER 24%	per gene DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_promoter_gene_type ARM CATEGORICAL 24%	per gene DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_promoter_host_tss_kb ARM REAL Â¥ 0 24%	Distance in kb from the precursor to its host's TSS (median 0.5 kb, max 874). AUTHORED
gctx_strand_rel ARM CATEGORICAL 27%	Is the precursor SENSE (633 arms) or ANTISENSE (40) to its host? AUTHORED

SURV_ Â· 8

reference â€” context, provenance and rarely-quoted detail

TCGA outcome panel legs.

8 columns Â·
 rung arm Â·
 coverage 6%â€”6% (median 6%) Â·
 5 share the block description

4x p-value Â·
 3x real â¥ 0 Â·
 1x constant

surv_dfi_hr_adj ARMREAL ± 06%	Adjusted DFI hazard ratio per arm “ range [0.84, 1.16] , median 1.00 . The whole distribution sits within ±16% of no effect. See <code>surv_dfi_q_adj</code> . AUTHORED Defined on: arms in the TCGA outcome panel (145) “ TCGA legs only; the GSE leg is bare-stem matched
surv_dfi_hr_base ARMREAL ± 06%	Unadjusted DFI hazard ratio for the arm (0.86 “ 1.16, median 1.00). AUTHORED
surv_dfi_p_adj ARMP-VALUE6%	p-value “ composition-adjusted DERIVED Defined on: arms in the TCGA outcome panel (145) “ TCGA legs only; the GSE leg is bare-stem matched
surv_dfi_p_base ARMP-VALUE6%	p-value DERIVED Defined on: arms in the TCGA outcome panel (145) “ TCGA legs only; the GSE leg is bare-stem matched
surv_dfi_q_adj ARMCONSTANT6%	BH q for the arm's DFI hazard ratio. AUTHORED
surv_os_hr_adj ARMREAL ± 06%	composition-adjusted DERIVED Defined on: arms in the TCGA outcome panel (145) “ TCGA legs only; the GSE leg is bare-stem matched
surv_os_p_adj ARMP-VALUE6%	p-value “ composition-adjusted DERIVED Defined on: arms in the TCGA outcome panel (145) “ TCGA legs only; the GSE leg is bare-stem matched
surv_os_q_adj ARMP-VALUE6%	BH-adjusted q “ composition-adjusted DERIVED Defined on: arms in the TCGA outcome panel (145) “ TCGA legs only; the GSE leg is bare-stem matched

AID_ “ 4

reference “ context, provenance and rarely-quoted detail

Arm-IDENTIFIABILITY rollup for the arm: how often this arm is separable from its same-seed mates (<code>arm_resolvable</code> , <code>arm_sep_z</code> , <code>oof_drho</code> medians). 4 columns “ rung arm “ coverage 6%“6% (median 6%) “ 3 share the block description 1× fraction 0“1 “ 1× real ± 0 “ 1× signed “’1“1 “ 1× count	
aid_frac_resolvable ARMFRACTION 0“1 AGG ARM-IN-FAMILY6%	fraction DERIVED Defined on: arms inside ““1 MULTI-ARM (gene,family) cell “ the arm rung's 20.3%-of-edges footprint (142)
aid_med_arm_sep_z ARMREAL ± 0 AGG ARM-IN-FAMILY6%	median “ per arm “ z-score DERIVED Defined on: arms inside ““1 MULTI-ARM (gene,family) cell “ the arm rung's 20.3%-of-edges footprint (142)
aid_med_oof_drho ARMSIGNED “’1“1 AGG ARM-IN-FAMILY6%	median “ out-of-fold DERIVED Defined on: arms inside ““1 MULTI-ARM (gene,family) cell “ the arm rung's 20.3%-of-edges footprint (142)
aid_n_cells ARMCOUNT AGG ARM-IN-FAMILY6%	Multi-arm cells this arm participates in “ fill 5.8% (the arm identifiability block is the sparsest on the card). AUTHORED

WSHIFT_ 6

reference “ context, provenance and rarely-quoted detail

Within-patient paired quantities on the ~103 matched tumour/NAT participants.

6 columns rung arm coverage 45%91% (median 52%) 6 share the block description

3x real 0 2x real, signed 1x count

wshift_mean_dGlobalRank ARMREAL, SIGNED52%	meanDERIVED Defined on: arms in the WITHIN-PATIENT paired subset (n% ^103 matched tumour/NAT pairs; 2,236 arms)
wshift_mean_own_shift ARMREAL, SIGNED52%	mean shiftDERIVED Defined on: arms in the WITHIN-PATIENT paired subset (n% ^103 matched tumour/NAT pairs; 2,236 arms)
wshift_n_pairs ARMCOUNT91%	countDERIVED Defined on: arms in the WITHIN-PATIENT paired subset (n% ^103 matched tumour/NAT pairs; 2,236 arms)
wshift_nat_patient_sd ARMREAL 074%	in matched normal (NAT) standard deviationDERIVED Defined on: arms in the WITHIN-PATIENT paired subset (n% ^103 matched tumour/NAT pairs; 2,236 arms)
wshift_own_specific_frac ARMREAL 045%	fractionDERIVED Defined on: arms in the WITHIN-PATIENT paired subset (n% ^103 matched tumour/NAT pairs; 2,236 arms)
wshift_sd_dGlobalRank ARMREAL 045%	standard deviationDERIVED Defined on: arms in the WITHIN-PATIENT paired subset (n% ^103 matched tumour/NAT pairs; 2,236 arms)

TIER_ 5

reference “ context, provenance and rarely-quoted detail

Expression tier of the arm against the detection floor “ class, fraction of samples expressed, p90 and max log2 RPM.

5 columns rung arm coverage 91%91% (median 91%) 2 share the block description

2x real 0 1x categorical 1x boolean 1x fraction 1

tier_class ARMCATEGORICAL91%	Expression tier “ silent on 1,518 of 2,236 arms (67.9%), conditional 476, the rest robust.AUTHORED
tier_expressed ARMBOOLEAN91%	Is the arm above the expression floor often enough to be usable (718 true / 1,518 false)?AUTHORED
tier_frac_expressed ARMFRACTION 191%	fractionDERIVED Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
tier_max ARMREAL 091%	maximumDERIVED Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
tier_p90 ARMREAL 091%	The arm's 90th-percentile RPM across tumour samples (0.10 “ 18.49, median 0.94).AUTHORED

SURROGATE_ 2

reference context, provenance and rarely-quoted detail

2 columns 1x fraction 0%1 1x categorical

2 columns 1x fraction 0%1 1x categorical

surrogate_corr

ARM FRACTION 0%1 0%

Correlation between this arm and its surrogate (0.62 0.89) the surrogate's quality gate. See the edge twin. AUTHORED

Defined on: arms with a same-seed SURROGATE (MH-166) 3.6% of edges

surrogate_instrument

ARM CATEGORICAL 0%

Which surrogate stands in for this arm's collapsed GTEx leg (4 levels). AUTHORED

DISC_ 2

reference context, provenance and rarely-quoted detail

2 columns 1x categorical 1x count

2 columns 1x categorical 1x count

disc_gold_families

ARM CATEGORICAL 1%

Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). AUTHORED

disc_n_gold_edges

ARM COUNT 100%

How many of the 157 discovery-queue edges name this arm. Concentrated: miR-106b-5p 39 miR-20a-5p 25 miR-93-5p 25 miR-19a-3p 19 miR-18a-5p 9. AUTHORED

FOC_ 7

reference context, provenance and rarely-quoted detail

7 columns 2x count 2x signed 1 1x identifier 1x boolean 1x p-value

7 columns 2x count 2x signed 1 1x identifier 1x boolean 1x p-value

foc_focal_locus

ARM IDENTIFIER 30%

The focal MIRBase locus used for this arm's CN analysis (437 distinct). AUTHORED

foc_multilocus

ARM BOOLEAN 30%

Does this arm map to MORE THAN ONE genomic locus (35 true / 698 false)? AUTHORED

foc_n

ARM COUNT 30%

count DERIVED

Defined on: arms with a paralogue-decontaminated FOCAL-locus concordance (733) DESCRIPTIVE, not an instrument

foc_n_loci

ARM COUNT 30%

count DERIVED

Defined on: arms with a paralogue-decontaminated FOCAL-locus concordance (733) DESCRIPTIVE, not an instrument

foc_partial_p

ARM P-VALUE 30%

p-value DERIVED

Defined on: arms with a paralogue-decontaminated FOCAL-locus concordance (733) DESCRIPTIVE, not an instrument

foc_partial_rho

ARM SIGNED 1 30%

Spearman r DERIVED

Defined on: arms with a paralogue-decontaminated FOCAL-locus concordance (733) DESCRIPTIVE, not an instrument

foc_spearman_rho

ARM

SIGNED $\hat{r}^2$

30%

Spearman $\hat{r}$

DERIVED

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) $\hat{r}$ DESCRIPTIVE, not an instrument

CNV_ 9

reference $\hat{r}$ context, provenance and rarely-quoted detail

Copy number at the arm's own locus $\hat{r}$ mean CN and amplification/deletion summaries.

9 columns $\hat{r}$ rung **arm** $\hat{r}$ coverage **35%** (median 35%) $\hat{r}$ 8 share the block description

6x percent $\hat{r}$ 3x real $\hat{r}$

cnv_mean_cn ARM REAL $\hat{r}$ 35%	mean DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) $\hat{r}$ multi-locus arms are DILUTED here
cnv_median_cn ARM REAL $\hat{r}$ 35%	median DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) $\hat{r}$ multi-locus arms are DILUTED here
cnv_pct_altered ARM PERCENT 35%	percent DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) $\hat{r}$ multi-locus arms are DILUTED here
cnv_pct_amp ARM PERCENT 35%	percent DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) $\hat{r}$ multi-locus arms are DILUTED here
cnv_pct_deep_del ARM PERCENT 35%	Percent of samples with a deep deletion at this arm's locus. AUTHORED
cnv_pct_gain ARM PERCENT 35%	percent DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) $\hat{r}$ multi-locus arms are DILUTED here
cnv_pct_loh ARM PERCENT 35%	percent DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) $\hat{r}$ multi-locus arms are DILUTED here
cnv_pct_loss ARM PERCENT 35%	percent DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) $\hat{r}$ multi-locus arms are DILUTED here
cnv_sd_cn ARM REAL $\hat{r}$ 35%	standard deviation DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) $\hat{r}$ multi-locus arms are DILUTED here

CNV_ 5

reference $\hat{r}$ context, provenance and rarely-quoted detail

CN-to-expression concordance fit for the arm (756 arms): does locus CN actually move this arm's abundance.

5 columns $\hat{r}$ rung **arm** $\hat{r}$ coverage **31%** (median 31%) $\hat{r}$ 4 share the block description

2x p-value $\hat{r}$ 2x signed $\hat{r}^2$ $\hat{r}$ 1x count

cnvc_n ARM COUNT 31%	count DERIVED Defined on: arms with a CN$\hat{r}$expression concordance fit (756)
---	---

cncv_partial_p ARM P-VALUE 31%	p-value DERIVED Defined on: arms with a CNâ€™’expression concordance fit (756)
cncv_partial_q ARM P-VALUE 31%	BH q for the arm's CN-coupling partial correlation. AUTHORED
cncv_partial_rho ARM SIGNED $\hat{A}^{'1}\hat{A}\epsilon 1$ 31%	Spearman $\hat{I}$ DERIVED Defined on: arms with a CNâ€™’expression concordance fit (756)
cncv_spearman_rho ARM SIGNED $\hat{A}^{'1}\hat{A}\epsilon 1$ 31%	Spearman $\hat{I}$ DERIVED Defined on: arms with a CNâ€™’expression concordance fit (756)

MODEL_ $\hat{A}$ 1

reference $\hat{A}$ ’’ context, provenance and rarely-quoted detail

1 columns $\hat{A}$ rung arm $\hat{A}$ coverage 30%$\hat{A}$’’30% (median 30%) 1x count
--

model_n_genes ARM COUNT AGG EDGE 30%	Which model variant produced this row. AUTHORED Defined on: arms carrying ≥ 1 curated HE edge in the model design (741)
---	---

COMV_ $\hat{A}$ 2

reference $\hat{A}$ ’’ context, provenance and rarely-quoted detail

2 columns $\hat{A}$ rung arm $\hat{A}$ coverage 15%$\hat{A}$’’15% (median 15%) 1x count $\hat{A}$ 1x identifier

comv_module ARM COUNT 15%	Which co-movement module this arm belongs to in the $\hat{A}$ §6b-RETIRED heuristic lane. AUTHORED
comv_top_partner ARM IDENTIFIER 15%	Co-expression module membership and top co-expressed partner. Pure co-expression $\hat{A}$ ’’ the cross-state class columns are deliberately excluded from this block. AUTHORED Defined on: arms in a co-expression module (371) $\hat{A}$’’ pure co-expression; the state-class columns are excluded

CLOAD_ $\hat{A}$ 3

reference $\hat{A}$ ’’ context, provenance and rarely-quoted detail

Compartment LOADING of the arm. 3 columns $\hat{A}$ rung arm $\hat{A}$ coverage 8%$\hat{A}$’’60% (median 60%) $\hat{A}$ 2 share the block description 1x real $\hat{A}\% \hat{0}$ $\hat{A}$ 1x fraction $\hat{0}\hat{A}$’’1 $\hat{A}$ 1x categorical
--

cload_dose_rpm ARM REAL $\hat{A}\% \hat{0}$ 60%	dose DERIVED Defined on: arms with a compartment-loading row (1,473) $\hat{A}$’’ $\hat{A}$’’ $\hat{A}$’’ its source has NO PRODUCER (MH-111 ad-hoc)
cload_lock ARM FRACTION $\hat{0}\hat{A}$’’1 8%	Is this arm's compartment loading LOCKED to one cell type rather than spread across the deconvolution? AUTHORED
cload_top_compartment ARM CATEGORICAL 60%	the top-ranked DERIVED Defined on: arms with a compartment-loading row (1,473) $\hat{A}$’’ $\hat{A}$’’ $\hat{A}$’’ its source has NO PRODUCER (MH-111 ad-hoc)

HX_ 1

reference " context, provenance and rarely-quoted detail

1	columns		run	arm		coverage	28%28%	(median 28%)
1x	real							

hx_hly_baseline_imputed

ARM	REAL 0	28%
-----	--------	-----

" IMPUTED healthy baseline (miTED rank-transfer or seed-mate substitution) " LABELLED
as imputed, never presented as a measurement. AUTHORED

seed_family_card.tsv

seed_family

One row per the seed family itself, gene-free. 83.7% of families are SINGLETONS for those every 'family-level' statistic is trivially that one arm's value. Mask on fam_single_member first.

33 columns 1,959 rows 2 blocks median coverage 92%

Contents 2 blocks, most important first

1. seed_1 CORE
2. fam_32 REFERENCE

SEED_1

core the columns findings rest on

1 columns 1 rung key 1 coverage 100% (median 100%)
1x identifier

seed_family

KEY the seed family itself, one row per family.

AUTHORED

KEY IDENTIFIER 100%

FAM_32

reference context, provenance and rarely-quoted detail

Seed-family structure the arm sits in membership, affinity and dominance within it.

32 columns 1 rung family 1 coverage 4% (median 91%) 16 share the block description
12x count 8x real 7x fraction 1 2x boolean 1x categorical 1x identifier 1x signed
1

fam_ctx_composition

FAMILY CATEGORICAL 29%

The family's genomic-context MIX as a tally string sense_coding_host:1 on 251 families, sense_lncRNA_host:1 on 175, antisense_overlap:1 next (29 distinct patterns).

AUTHORED

fam_ctx_context_homogeneous

FAMILY BOOLEAN 29%

Do ALL of this family's members share one genomic context True on 524 of 571 (91.8%)?

AUTHORED

fam_ctx_dose_coding_host_frac

FAMILY FRACTION 1

29%

dose fraction DERIVED

Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)

fam_ctx_dose_cotranscribed_frac

FAMILY FRACTION 1

29%

dose fraction DERIVED

Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)

fam_ctx_n_distinct_coding_hosts

FAMILY COUNT 29%

count DERIVED

Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)

fam_cur_degree_max FAMILY COUNT 36%	maximum DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_dominant_arm FAMILY IDENTIFIER 92%	the dominant one $\hat{A}$ · per arm DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_dominant_share FAMILY FRACTION $\hat{O}\hat{A}\hat{\epsilon}$"1 92%	The top member's share of the family's total dose. AUTHORED
fam_dose_hhi FAMILY FRACTION $\hat{O}\hat{A}\hat{\epsilon}$"1 15%	Concentration of DOSE across the family's members. $\hat{a}\epsilon$... FLOOR-CORRECTED $\hat{a}\epsilon$" verified 2026-08-19 : its minimum sits far below the raw 1/k floor at every member count (0.0000 at k=2 where the raw floor is 0.500), and spearman(n_members, hhi) = +0.1896 rather than the strongly negative value a raw index would give. AUTHORED
fam_dose_max_log2 FAMILY REAL $\hat{A}\hat{\%}\hat{Y}$ 0 92%	dose $\hat{A}$ · maximum DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_dose_med_log2 FAMILY REAL $\hat{A}\hat{\%}\hat{Y}$ 0 92%	dose $\hat{A}$ · median DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_dose_total_log2 FAMILY REAL $\hat{A}\hat{\%}\hat{Y}$ 0 100%	dose $\hat{A}$ · total DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_fame_npmid_sum FAMILY COUNT 100%	sum DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_n_expressed FAMILY COUNT 100%	count DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_n_guide FAMILY COUNT 100%	count DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_n_members FAMILY COUNT 100%	count $\hat{A}$ · family members DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_n_members_context FAMILY COUNT 29%	count $\hat{A}$ · family members DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_n_model_edges FAMILY COUNT 100%	count DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_n_passenger FAMILY COUNT 100%	count DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)
fam_n_shift_gt20_gated FAMILY COUNT 100%	count $\hat{A}$ · shift $\hat{A}$ · gated DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{a}$' 83.7% single-member $\hat{a}$*' shares/HHI are 1 or NaN)

fam_real_n_c10_sum FAMILY COUNT 100%	count · at the 0.10 threshold · sum DERIVED Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)
fam_real_n_scored_sum FAMILY COUNT 100%	count · sum DERIVED Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)
fam_redund_m_eff FAMILY REAL % 0 4%	EFFECTIVE number of independent members after accounting for within-family correlation · the family's redundancy-corrected size (mode 2.0, 50 distinct values). AUTHORED
fam_redund_median_within_rho FAMILY SIGNED -11 4%	median · Spearman ρ DERIVED Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)
fam_redund_overcount_frac FAMILY FRACTION 1 4%	fraction DERIVED Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)
fam_seed_shift_max_gated FAMILY FRACTION 1 13%	shift · maximum · gated DERIVED Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)
fam_single_member FAMILY BOOLEAN 100%	· THE DOMINANT FACT ABOUT THIS ENTIRE CARD: 1,639 of 1,959 families (83.7%) are SINGLETONS. For those rows every 'family-level' statistic is TRIVIALY that one arm's value, and <code>fam_dominant_share</code> is forced to exactly 1.0. AUTHORED
fam_targetome_seed_med FAMILY REAL % 0 97%	median DERIVED Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)
fam_targetome_seq_med FAMILY REAL % 0 30%	median DERIVED Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)
fam_var_hhi FAMILY FRACTION 1 14%	Herfindahl concentration of <code>abund_sd</code> ACROSS the family's members · is the family's variance carried by one arm? AUTHORED
fam_var_max FAMILY REAL % 0 89%	maximum DERIVED Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)
fam_var_med FAMILY REAL % 0 89%	median DERIVED Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)